INTRINSIC GRADIENT BARRIERS IN PUSHFORWARD DEEP GENERATIVE MODELS

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Broad classes of deep generative models (DGMs) operate by learning to map a simple source distribution, such as a standard Gaussian, to a complex high-dimensional target distribution. Despite their success, DGMs often empirically struggle to accurately represent distributions with heavy tails or well-separated modes. In this work, we show that these difficulties reflect intrinsic properties of the underlying transport problem rather than particular neural network architectures or training procedures. We establish quantitative lower bounds on the gradients and Jacobians of pushforward mappings that transport log-concave source distributions to such targets. For heavy-tailed target distributions, the required gradient magnitude grows as the target develops increasingly large moments. For multimodal target distributions, the low-probability regions between well-separated modes similarly force large gradients in the transport map. We discuss how such large derivatives relate to trade-offs between expressivity and numerical stability. Our results hold under substantially weaker regularity conditions than previous analyses and remain stable under perturbations of the target distribution. Finally, we show that these obstructions can be mitigated when the goal is approximation of a given density rather than generative modeling. Specifically, we propose simple classes of bijective reparametrizations that can alleviate the difficult geometric structure and reduce the gradient burden placed on the learned generative map. Numerical experiments support our theoretical analysis.

1. Introduction. Deep generative models (DGMs) are foundational tools in modern artificial intelligence. Despite their success in tasks such as image generation, it has been empirically observed that DGMs often struggle when the target distribution is heavy-tailed [37, 54] or multimodal [59, 67]. In particular, DGMs can grossly under-characterize tail events [66] and fail to represent distinct modes [36]. These phenomena are consequential in practice since DGMs are increasingly adopted in scientific domains where heavy-tailed or multimodal data are the norm rather than the exception [32, 46]. Heavy-tailed data arise in applications such as the climate sciences [16, 35], where inadequate representation of extreme events can lead to under-estimation of extreme climate occurrences [75], and finance [9], where underestimating tail probabilities can distort estimates of important risk measures such as Expected Shortfall [19, 50]. Similarly, multimodal distributions arise in applications ranging from single-cell analysis, where missing modes can obscure biologically distinct cell populations [48], to the modeling of heterogeneous populations, where omitting modes associated with under-represented subgroups can undermine the accuracy and fairness of downstream analyses [47].

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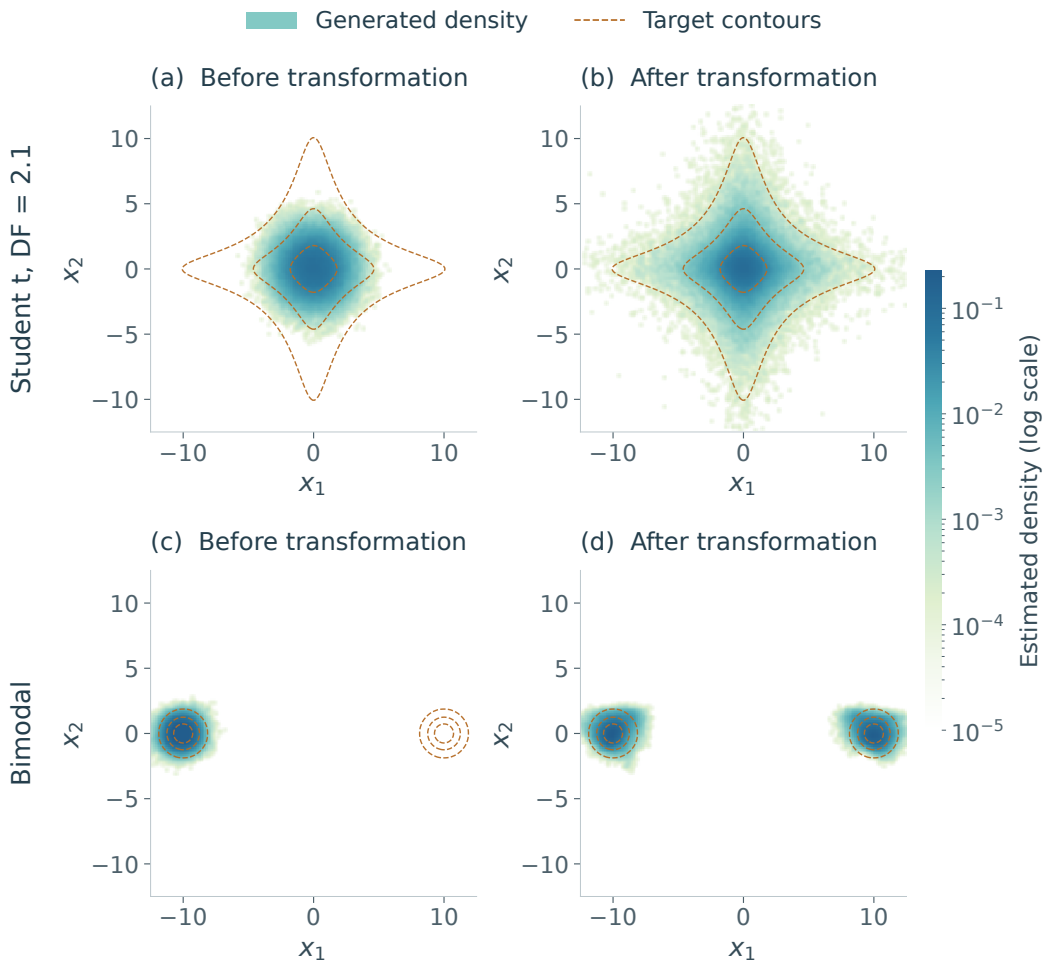


Fig 1: Results from using normalizing flows with a Gaussian source distribution to approximately sample from two target densities before (**left two panels**) and after (**right two panels**) applying bijective reparametrizations. The targets are a heavy-tailed Student-t distribution (top) and a bimodal Gaussian mixture (bottom). Heatmaps show the densities estimated from the generated samples in the original coordinates, while dashed curves show the target contours.

As DGMs have become more widely used, it is critical to understand *why* DGMs can struggle to faithfully represent heavy-tailed and multimodal distributions. In this paper, we study the *intrinsic* limitations of pushforward DGMs in representing such distributions. DGMs can be viewed through the lens of learning a map $T : \mathcal{Z} \to \mathcal{X}$ that pushes a simple source distribution ν (most commonly a Gaussian) on some latent space $\mathcal{Z}$ to a target distribution μ on the sample space $\mathcal{X}$. Prior work has examined limitations arising from global Lipschitz constraints on the neural network map $\widehat{T}$ used to approximate T [34, 59, 66]. We ask a more fundamental question: what regularity constraints does the target distribution μ itself impose on *any* transport map T satisfying $T_{\#}\nu = \mu$? This perspective isolates the intrinsic obstruction in transporting ν to μ, independently of the architecture or estimation procedure used to approximate the map. As a result, our analysis applies to a broad class of models, including various normalizing flow models whose induced transport maps need not be globally Lipschitz.

The left two panels of Figure 1 highlight the intrinsic difficulties of implicitly sampling from two challenging target densities (dashed contours): a Student-t with 2.1 degrees of freedom (top) and a bimodal Gaussian mixture (bottom). As shown by the heatmaps of the approximate samples, normalizing flows under a Gaussian source distribution struggle significantly with these targets. Specifically, they exhibit severe tail underestimation for the Student-t density in Figure 1(a) and complete omission of one of the modes for the Gaussian mixture in Figure 1(c). The experimental details for these simulations are provided in Section S5.5 of the Supplement.

For our theoretical analysis of heavy-tailed targets, we leverage Poincaré-type inequalities to show that under the predominantly used Gaussian source measures, pushforward maps to targets with large centered moments must have large gradient components and Jacobians. We further generalize these results to log-concave and strongly log-concave source distributions. For multimodal target distributions with well-separated modes, we use isoperimetric and coarea inequalities to show that large gradients are a mathematical necessity, even when the target's moments remain bounded. It is well-known that, for invertible generative models, large transport derivatives can amplify numerical errors [4], leading to a tradeoff between expressivity and numerical stability. Our results thus unveil insights into why common DGMs can fundamentally struggle to learn transport maps for heavy-tailed and/or multimodal targets, *irrespective* of their underlying neural network parameterization.

Our analysis also suggests ways to mitigate these obstructions when the goal is to use such neural networks for approximation of a given target density rather than generative modeling. The target distribution in generative modeling is typically specified by training samples [52]. When the target density is known up to a normalizing constant, as is typical in variational inference [56], we can leverage this additional information to mitigate these obstructions. The key idea is to use known bijective reparametrizations to absorb part of the required derivative growth, leaving a more regular map to be learned. Although the resulting composite map remains subject to our lower bounds, this decomposition reduces the burden on the learned component of the map. The right two panels of Figure 1 demonstrate the practical benefits of applying these bijective reparameterizations. Specifically, Figure 1(b) shows that the tail regions of the heavy-tailed Student-t density are better represented, while Figure 1(d) demonstrates that both modes of the Gaussian mixture are correctly captured.

Our main contributions are summarized as follows:

1. For heavy-tailed targets, we derive lower bounds on transport gradients and Jacobians in terms of centered target moments. Our theory holds under Gaussian and log-concave source distributions without assuming a global Lipschitz bound. Importantly, these results characterize intrinsic limitations of the underlying transport problem, independent of any neural network architecture or training procedure.
2. For multimodal targets, we similarly show that low-probability valleys between modes force large transport gradients and Jacobians.
3. When the goal is to approximate a given density, we propose simple bijective reparametrizations for heavy-tailed and multimodal targets that reduce the derivative burden on the learned map. This allows us to alleviate the intrinsic gradient barriers in pushforward DGMs and more faithfully represent these challenging densities.

The remainder of the paper is organized as follows. Section 2 reviews related work, and Section 3 introduces the notation and setup. Section 4 uses functional inequalities to derive gradient lower bounds in terms of target moments, while Section 5 establishes localized Jacobian bounds for multimodal targets. Section 6 discusses practical implications for DGM architecture design and numerical stability. Section 7 develops bijective reparametrizations for density approximation, and Section 8 presents numerical illustrations. Proofs, technical background, and additional experiments are provided in the Supplement.

2. Related Work. In this work, we focus on pushforward-based deep generative models, where the generated distribution is realized as the pushforward $T_{\#}\nu$ of a fixed source distribution ν by a learned map T. This perspective encompasses models that represent such maps explicitly with neural networks, including generative adversarial networks (GANs; [28]), variational autoencoders (VAEs; [38]), and normalizing flows [17, 56]. It also includes continuous flow models, such as neural ordinary differential equations (ODEs) and related stochastic flow models [1, 12, 44], where a learned vector field is integrated over time to produce a transport map. Diffusion models [31, 63, 64] can also be treated within this pushforward framework after time discretization; the reverse sampling procedures can be viewed as pushforwards of a Gaussian measure on an augmented space [59, 66]. Our results therefore apply to the resulting discretized sampling maps. Models that do not naturally specify a pushforward mechanism, such as energy-based models defined by unnormalized densities or large language models based on attention architectures, fall outside the scope of our analysis.

Several prior works have quantified the expressive power of DGMs under light-tailed/moments assumptions on the target [45, 74]. However, there is growing empirical and methodological evidence that standard DGMs struggle to faithfully represent heavy-tailed targets [26, 34, 66, 72]. These observations have motivated work aimed at mitigating such deficiencies, such as heavy-tailed diffusions [54], Pareto GANs [33], and tail-adaptive normalizing flows [30]. For multimodal targets, [59] and [67] relate mode separation and approximation error to global Lipschitz constraints. Our multimodal analysis yields lower bounds on derivative moments under only local Lipschitz regularity.

The most closely related prior works are [34] and [59], who study heavy-tailed and multimodal targets under global Lipschitz assumptions. Related analyses of invertible networks and bi-Lipschitz flows also use global Lipschitz bounds [29, 70]. Our results instead constrain gradients and Jacobians directly, without imposing a global bound on their size and allowing transport maps with unbounded derivatives. We require only Sobolev regularity for the moment bounds and local Lipschitz continuity for the multimodal bounds. These conditions are substantially weaker and cover a broader class of generative models, including normalizing flows and continuous flows whose transport maps need not be globally Lipschitz.

The main tools that we use in our analysis are functional and isoperimetric inequalities. These inequalities play a central role in probability theory, functional analysis, and the study of Markov chains [2]. In the context of sampling, such inequalities are often used to analyze the convergence of sampling algorithms such as Langevin dynamics [15]. While these inequalities are typically used to derive upper bounds on a Markov chain's distance to stationarity in the sampling literature, here we use them to *lower-bound* the size of transport gradients and Jacobians.

The gradient bounds derived in our analysis motivate changing the coordinates in which a target density is approximated. Such coordinate transformations have a long history in the sampling literature. For example, measure transport methods [49] and normalizing flows [25, 53] have been used to construct changes of coordinates that facilitate Monte Carlo sampling from complex target distributions. Gaussianization methods [13, 27, 58] seek transformations that make a target distribution approximately Gaussian, whereas stereographic MCMC [73] uses stereographic projections to reformulate the sampling problem on a sphere. In this article, we develop simple, bijective reparametrizations specifically for heavy-tailed and multimodal target densities.

3. Preliminaries.

3.1. *Setup and Definitions.* For a matrix $\mathbf{A}$, $\|\mathbf{A}\|_F$ denotes its Frobenius norm, and $\|\mathbf{A}\|_{\mathrm{op}}$ denotes its operator norm. For a vector $x \in \mathbb{R}^d$, $\|x\|_2$ denotes the Euclidean norm. Generic positive constants are denoted by C, c, whose values may change from line to line.

Probability measures are denoted by Greek letters (e.g., ν, μ), with γ reserved for the standard Gaussian measure. For a measurable map $T : \mathcal{Z} \to \mathcal{X}$ and a probability measure ν on $\mathcal{Z}$, we write $T_{\#}\nu$ for the pushforward measure induced by T. We denote by $J_T(x)$ the Jacobian of the map T at x, and by ∇f the gradient of a scalar-valued function f. For vector-valued maps $T = (T_1, \dots, T_m)$, the Jacobian J_T has entries $(J_T)_{ij} = \partial T_i / \partial x_j$. As is standard in the literature, when we write $\mathrm{Var}_\nu(f)$ or $\mathbb{E}_\nu(f)$, we mean $\mathrm{Var}(f(Z))$ and $\mathbb{E}(f(Z))$ for $Z \sim \nu$. Following a convention in high dimensional geometry, we call ν isotropic if it is centered and has identity covariance. We use κ for Student-t degrees of freedom, α for the Pareto shape parameter, and ζ for the Fréchet shape parameter.

Let $1 \le p < \infty$. For a full-dimensional log-concave source ν, write $\Omega = \mathrm{int}(\mathrm{supp}\, \nu)$. A function f belongs to the Sobolev space $W^{1,p}(\nu)$ if $f \in W^{1,p}_{\mathrm{loc}}(\Omega)$ and both f and its weak gradient ∇f belong to $L^p(\nu)$. Weak derivatives are taken on Ω; values outside Ω are irrelevant. All gradients and integrals in this work are understood in the above sense. See Section S1 of the Supplement for background on weak gradients and Sobolev spaces.

A random variable $X \in \mathbb{R}$ is said to be heavy-tailed if $\mathbb{E}[e^{t|X|}] = \infty$ for all $t > 0$. A fundamental link between the pth moment of X (if it exists) and its tail probabilities is given by the polynomial Markov (Chebyshev) inequality: for any $a > 0$, $\mathbb{P}(|X| \ge a) \le \mathbb{E}|X|^p / a^p$. This inequality shows that finite pth moments imply at least polynomial decay of the tail. We give more technical details on moments and tails in Section S1 of the Supplement.

3.2. *Pushforward Maps and Deep Generative Models.* Throughout, we adopt the pushforward perspective on DGMs. Let ν be a probability measure on a latent space $\mathcal{Z} \subseteq \mathbb{R}^d$ and let μ be a target probability measure on $\mathcal{X} \subseteq \mathbb{R}^m$. A measurable map $T : \mathcal{Z} \to \mathcal{X}$ is said to push ν forward to μ if $T_{\#}\nu = \mu$, i.e.,

$$\mu(A) = \nu\big(T^{-1}(A)\big) \quad \text{for all Borel } A \subseteq \mathcal{X}.$$

When both μ and ν are non-atomic, the existence of such maps is guaranteed, although they are generally not unique [71].

In DGMs, one learns an approximation to such a pushforward map. That is, one considers a family $\{T_\theta : \theta \in \Theta\}$ of functions (typically involving neural networks) and seeks parameters $\hat{\theta}$ such that $(T_{\hat{\theta}})_{\#}\nu \approx \mu$ in an appropriate sense. The most typical setting in DGMs is to set $\nu = \mathcal{N}(0, I_d)$. One then generates approximate samples from μ by transforming a sample $Z \sim \nu$ into $X = T_{\hat{\theta}}(Z)$.

In models such as GANs and VAEs, the mapping T_θ is modeled explicitly by a neural network, where training involves minimizing a divergence (e.g., Jensen-Shannon or Kullback-Leibler) between μ and $(T_\theta)_{\#}\nu$. In normalizing flows where the map is required to be invertible, neural networks with special coupling layers that ensure invertibility are used to model T_θ. In dynamic models such as continuous flows, T_θ is realized implicitly through a composition of infinitesimal transformations (e.g., vector fields) that are modeled by neural networks. See [69] for a comprehensive overview of DGMs.

4. Gradient Bounds for Deep Generative Models via Functional Inequalities. Our goal is to formalize the principle that large moments of the target distribution necessarily require large gradients or Jacobians of the transport map, with an appropriate notion of size. To this end, it is natural to consider Poincaré inequalities, which relate the variance of the target to the gradient of the pushforward map. We begin with the most relevant case when the source distribution is Gaussian. Proofs of the exact-transport results in this section are delegated to Section S2 of the Supplement.

4.1. *Gaussian Source Distributions.* We say that ν satisfies a Poincaré inequality with constant $C_{\mathrm{PI}}(\nu) > 0$ if for all functions $f : \mathbb{R}^d \to \mathbb{R}$ belonging to $W^{1,2}(\nu)$ with finite variance, the following bound holds:

$$\mathrm{Var}_\nu(f) \le C_{\mathrm{PI}}(\nu)\, \mathbb{E}_\nu[\|\nabla f\|_2^2]. \tag{4.1}$$

A Poincaré inequality implies that if $f(Z)$ has high variance (i.e., if $f_{\#}\nu$ has a large centered second moment), then the gradient ∇f must be large in expectation. In particular, the standard Gaussian measure $\mathcal{N}(0, I_d)$ satisfies (4.1). Further background on the Poincaré constant is provided in Section S1.6 of the Supplement.

LEMMA 4.1 (Gaussian Poincaré inequality). *Let γ denote the standard Gaussian measure on $\mathbb{R}^d$, and let $f \in W^{1,2}(\gamma)$. Then*

$$\mathrm{Var}_\gamma(f) \le C \int_{\mathbb{R}^d} \|\nabla f(x)\|_2^2 \, d\gamma(x), \tag{4.2}$$

where $C = 1$.

Intuitively, the Gaussian Poincaré inequality implies that producing a target distribution with large variance from a Gaussian source requires a pushforward map whose gradient has a large expected squared norm.

The space $W^{1,2}(\gamma)$ in Lemma 4.1 only requires very weak regularity conditions (weak differentiability of the function and square-integrability of its gradient) that ensures expectations and gradients are well defined. This is substantially more general than global Lipschitz assumptions, which impose a uniform upper bound on the gradient. Notably, the inequality (4.2) is independent of the input dimension. It is a well-known fact that inequality (4.2) is tight, with equality achieved by affine functions [41].

Since DGMs concern mappings from a latent space to a high-dimensional target space, this motivates a multivariate extension of the Gaussian Poincaré inequality.

THEOREM 4.2 (Gaussian Poincaré inequality for vector-valued maps). *Let $Z \sim \mathcal{N}(0, I_d)$ and let $T : \mathbb{R}^d \to \mathbb{R}^m$ have components $T = (T_1, \ldots, T_m)$ with $T_j \in W^{1,2}(\gamma)$ for each j. Let $J_T(z) \in \mathbb{R}^{m\times d}$ denote the Jacobian of T. Then:*

(i) Component-wise variance bounds *For each $j = 1, \ldots, m$,*

$$\mathrm{Var}\big(T_j(Z)\big) \le \mathbb{E}\big[\|\nabla T_j(Z)\|_2^2\big].$$

(ii) Covariance matrix bound *For $\mathrm{Cov}(T(Z)) \in \mathbb{R}^{m\times m}$, the covariance matrix of $T(Z)$, we have the following inequality:*

$$\mathrm{Cov}\big(T(Z)\big) \preceq \mathbb{E}\big[\, J_T(Z)\, J_T(Z)^\top\big],$$

where $\preceq$ denotes the Loewner order on the positive semidefinite matrices. Equivalently, for every $a \in \mathbb{R}^m$,

$$\mathrm{Var}\big(a^\top T(Z)\big) \le \mathbb{E}\big[\|J_T(Z)^\top a\|_2^2\big] = a^\top \mathbb{E}\big[J_T(Z) J_T(Z)^\top\big] a.$$

(iii) Trace Bound *In particular,*

$$\mathrm{Tr}\,\big[\mathrm{Cov}(T(Z))\big] \le \mathbb{E}\big[\|J_T(Z)\|_F^2\big] = \sum_{j=1}^m \mathbb{E}\big[\|\nabla T_j(Z)\|_2^2\big].$$

(iv) Entrywise covariance bound *For $i, j \in \{1, \ldots, m\}$,*

$$|\mathrm{Cov}(T_i(Z), T_j(Z))| \le \sqrt{\mathbb{E}\|\nabla T_i(Z)\|_2^2}\,\sqrt{\mathbb{E}\|\nabla T_j(Z)\|_2^2}.$$

The above inequalities relate the covariance of $T(Z)$ with the gradients and Jacobians of the map T. As an immediate consequence, if the target $\mu = T_{\#}\nu$ exhibits large centered second moments in its components, the Jacobian of the pushforward map T must be correspondingly large. We discuss the practical implications of these results for DGMs in Section 6.

4.2. *L_p–L_q Poincaré Inequalities.* The Gaussian Poincaré inequality lower bounds the expected gradient components and Jacobian by the second moments of the target. Given the close relationship between higher-order moments and heavy-tailed behavior, it is natural to seek analogous lower bounds on the gradients and Jacobian of the pushforward map in terms of higher-order moments of the target distribution. To achieve this, we use more recently developed mathematical tools known as L_p–L_q Poincaré inequalities [14, 51, 62]. An L_p–L_q Poincaré inequality for a probability measure ν on $\mathbb{R}^d$ asserts the existence of a positive constant $C_{p,q} < \infty$ such that

$$\|f - \mathbb{E}_\nu f\|_{L^p(\nu)} \leq C_{p,q} \big\| \|\nabla f\|_2 \big\|_{L^q(\nu)}, \tag{4.3}$$

for all $f \in W^{1,q}(\nu) \cap L^p(\nu)$ for $p, q \in [1, \infty]$ with $q \geq p$. In other words, (4.3) quantifies how the pth centered moment of the target lower bounds the L^q-norm of the pushforward's gradient. The classical Poincaré inequality corresponds to the case $p = q = 2$. For log-concave ν, there is a natural notion of equivalence between L_p–L_q Poincaré inequalities with different p's and q's [14, 51]. For our purposes, we will focus on the case where $p = q$, and refer to these as L_p Poincaré inequalities.

It is well known that the standard Gaussian measure satisfies an L_p Poincaré inequality for all $2 \leq p < \infty$, with constant C_p that depends on p but not on d.

LEMMA 4.3 (L_p Gaussian Poincaré inequalities [7]). *Let γ denote the standard Gaussian measure on $\mathbb{R}^d$ and let $Z \sim \mathcal{N}(0, I_d)$. For any $2 \leq p < \infty$ there exists a finite constant C_p depending only on p such that for all $f \in W^{1,p}(\gamma)$,*

$$\|f - \mathbb{E}_\gamma f\|_{L^p(\gamma)} \leq C_p \big\| \|\nabla f\|_2 \big\|_{L^p(\gamma)}.$$

Lemma 4.3 provides a lower bound on the norm of the gradient of a scalar-valued pushforward map in terms of the higher centered moments of the target distribution. Since DGMs have high-dimensional targets, we develop the corresponding multivariate bound below.

COROLLARY 4.4 (L_p Gaussian Poincaré for vector-valued maps). *Let $T = (T_1, \ldots, T_m) : \mathbb{R}^d \to \mathbb{R}^m$ with $T_j \in W^{1,p}(\gamma)$ for some $p \geq 2$. We have*

$$\sum_{j=1}^{m} \|T_j - \mathbb{E}_\gamma T_j\|^p_{L^p(\gamma)} \leq C'_p \, \mathbb{E}_\gamma \|J_T\|^p_F.$$

where $C'_p > 0$ is some constant that depends on p.

An immediate consequence of Corollary 4.4 is that if the target distribution exhibits large centered pth moments $\mathbb{E}|T_j(Z) - \mathbb{E}T_j(Z)|^p$ for one or more components, then the expected pth power of the Jacobian norm must be correspondingly large.

4.3. *More General Log-Concave Distributions.* A natural question is whether the above results hold only for Gaussian measures. It turns out that similar results also hold for the more general class of log-concave measures $\nu \propto e^{-V}$ for V convex. These include the Laplace and logistic distributions, uniform distributions on convex bodies, exponential distributions, and

gamma distributions with shape parameter of at least one. Some of these distributions have been explored as alternative latent priors in DGMs [24]. Definitions of log-concavity and strong log-concavity are provided in Section S1.4 of the Supplement.

In the strongly log-concave regime where V satisfies the lower bound $\nabla^2 V \succeq t \cdot I_d$ for $t > 0$, it is known that $C_{\mathrm{PI}}(\nu) \leq 1/t$ [2, 41], implying Gaussian-like Poincaré inequalities. When such a curvature lower bound is not available, it is known that isotropic log-concave measures satisfy the Poincaré inequality with constant $C_{\mathrm{PI}}(\nu) \leq C \log(ed)$ [6, 39]. As such, the dimension dependence is only logarithmic. If a conjecture known as the Kannan-Lovász-Simonovits (KLS) conjecture holds true, then even this $\log(ed)$ factor may be removed [40, 43]. Hence, our results hold analogously for not just the typical case of Gaussian source distributions, but also for general log-concave measures.

In our setting, these results imply that all moment-to-gradient lower bounds derived from Poincaré or $L_p - L_q$-Poincaré inequalities extend verbatim to log-concave sources but with constants that are dependent on ν. Concretely, we have the following theorem.

THEOREM 4.5 (Log-concave Poincaré inequalities [6, 39]). *Let ν be an isotropic log-concave measure on $\mathbb{R}^d$. Then for all smooth $f \in W^{1,2}(\nu)$, we have*

$$\mathrm{Var}_\nu(f) \leq C_{\mathrm{PI}}(\nu) \mathbb{E}_\nu \|\nabla f\|_2^2,$$

where $0 < C_{\mathrm{PI}}(\nu) \leq C \log(ed)$ for a universal constant $C > 0$.

Similar results also extend to the $L_p - L_q$ case. We state the case for $p = q$ below.

THEOREM 4.6 (L_p-Poincaré inequalities for isotropic log-concave measures [51]). *Let ν be an isotropic log-concave measure on $\mathbb{R}^d$. Then for all smooth $f \in W^{1,p}(\nu)$ where $2 \leq p < \infty$, we have*

$$\|f - \mathbb{E}_\nu f\|_{L^p(\nu)} \leq C_{p,d,\nu} \big\| \|\nabla f\|_2 \big\|_{L^p(\nu)}.$$

where $C_{p,d,\nu} > 0$ is a constant that can depend on ν, p and d.

The proof for $L_p - L_q$-Poincaré inequalities for log-concave measures is first found in the seminal work of [51], and an exposition is in [14]. Thus, our results and analysis for the case of a Gaussian source distribution follow through to the much more general case of a log-concave source. In this scenario, heavy-tailed targets with large moments once again force large Jacobian norms of any pushforward map.

4.4. *Approximate Pushforward Maps.* Our previous bounds concern an exact pushforward map. In practice, however, a learned model only approximates the target. It is therefore useful to ask whether the same conclusion continues to hold for an approximation. For the second-moment bound, Wasserstein distance provides a direct answer. For measures P, Q with finite second moments, define

$$W_2(P, Q) = \inf \big\{ (\mathbb{E}\|X - Y\|_2^2)^{1/2} : X \sim P,\ Y \sim Q \big\},$$

where the infimum is over couplings of the two measures.

PROPOSITION 4.7 (Stability of the second-moment bound). *Let T have components in $W^{1,2}(\gamma)$, let $\widehat{\mu} = T_{\#}\gamma$, and suppose that the target μ has finite second moments. If $W_2(\widehat{\mu}, \mu) \leq \varepsilon$, then*

$$\big(\mathbb{E}_\gamma \|J_T\|_F^2\big)^{1/2} \geq \big(\sqrt{\mathrm{Tr}[\mathrm{Cov}(\mu)]} - \varepsilon\big)_+,$$

where $(a)_+ = \max(a, 0)$.

Proposition 4.7 shows that an approximation in W_2 *cannot* eliminate the gradient requirement when the error is small relative to the square root of the target's centered second moment. The proof of Proposition 4.7 is given in Section S2 of the Supplement.

5. Gradient Bounds for Multimodal Targets. While moment bounds quantify the derivative growth associated with heavy-tailed targets, they do not generally capture the difficulty of transporting to well-separated modes. Large moments in the target distribution are not the only source of large Jacobians. In this section, we show that accurate representation of target distributions with well-separated modes *also* requires large Jacobians.

Here, the relevant property is the presence of two regions with substantial probability separated by a low-probability region, which we refer to as a valley. An accurate transport map must send substantial probability mass to each of the two high probability regions while sending little mass to the valley. In what follows, we use isoperimetric inequalities to show that satisfying these requirements forces large Jacobian norms on the transport map, with quantitative bounds determined by the separation and the probability masses of these regions. This generalizes and complements analyses that rely on global Lipschitz assumptions [59, 67]. All proofs in this section are given in Section S3 of the Supplement.

5.1. *A Localized Jacobian Bound.* Let γ be the standard Gaussian measure on $\mathbb{R}^d$, and let $T : \mathbb{R}^d \to \mathbb{R}^m$ be locally Lipschitz. We write $P = T_{\#}\gamma$ for the learned distribution. Local Lipschitz continuity ensures that the Jacobian J_T exists almost everywhere, but we impose no uniform bound on its norm. Here, $\|J_T\|_{\mathrm{op}}$ denotes the operator norm induced by the Euclidean norms.

For a nonempty closed set $A \subseteq \mathbb{R}^m$, define

$$A^t = \{y : \mathrm{dist}(y, A) \le t\}, \qquad U = \{y : 0 < \mathrm{dist}(y, A) < \Delta\}, \tag{5.1}$$

where $\Delta > 0$ and the set U separates A from the complement of A^Δ. We will apply the bound when the target assigns substantial probability to both A and the complement of A^Δ but little probability to U. Let φ and Φ denote the density and cumulative distribution function (CDF) of a standard univariate Gaussian. Define the Gaussian isoperimetric profile function as $I(s) = \varphi(\Phi^{-1}(s))$ for $0 < s < 1$, with $I(0) = I(1) = 0$. Intuitively, the Gaussian isoperimetric profile $I(s)$ quantifies the smallest possible size of the boundary of a region with probability s. It expresses the fact that dividing a Gaussian distribution into two regions of substantial probability requires a boundary of non-negligible size. It is a known mathematical fact that this function is symmetric about $1/2$ and increasing on $[0, 1/2]$ [51]. A general definition and further interpretation of isoperimetric profiles are given in Section S1.5 of the Supplement.

THEOREM 5.1 (Localized Jacobian bound). *Let $T : \mathbb{R}^d \to \mathbb{R}^m$ be locally Lipschitz and $P = T_{\#}\gamma$. For every nonempty closed $A \subseteq \mathbb{R}^m$ and every $\Delta > 0$, with U as in (5.1),*

$$\int_{T^{-1}(U)} \|J_T(z)\|_{\mathrm{op}}\, d\gamma(z) \ge \int_0^\Delta I\big(P(A^t)\big)\, dt \ge \Delta \cdot I(b), \tag{5.2}$$

where $b = \min\{P(A), 1 - P(A^\Delta)\}$.

Note that no integrability assumption on J_T is required; the integral on the left-hand side may be infinite. Theorem 5.1 identifies where the map must have large Jacobian norms. Intuitively, $I(b)$ gives the minimum size of a boundary needed to partition the source space into two regions each with probability mass at least b. We therefore require a contribution of at least $\Delta \cdot I(b)$ to the Jacobian from the points mapped into the valley U. For fixed b, this lower bound increases linearly with the separation Δ.

We stress that our results do not depend on the dimension of the source or target distributions. The proof of Theorem 5.1 applies Gaussian isoperimetry to the level sets of the distance from $T(z)$ to A and integrates the result over distances from 0 to Δ. Proposition 5.3 extends these localized and approximation bounds to general log-concave sources, with a source-dependent constant.

5.2. *Approximation of Multimodal Targets.* We next apply the localized Jacobian bound (5.2) in Theorem 5.1 to maps that approximate a target distribution μ. Define the total variation distance by $\mathrm{TV}(P,\mu) = \sup_B |P(B) - \mu(B)|$, where the supremum is over Borel sets. Suppose that for the sets in (5.1),

$$\mu(A) \geq w_1 > 0, \qquad \mu\big((A^{\Delta})^c\big) \geq w_2 > 0, \qquad \mu(U) \leq \delta. \tag{5.3}$$

These conditions describe the geometry needed for the result without requiring a density or a mixture representation.

COROLLARY 5.2 (Jacobian moments for a small-probability valley). *Suppose that* (5.3) *holds, and write* $w = \min\{w_1, w_2\}$. *Let* $T : \mathbb{R}^d \to \mathbb{R}^m$ *be locally Lipschitz, with* $P = T_{\#}\gamma$ *and* $\mathrm{TV}(P,\mu) \leq \varepsilon < w$. *Then*

$$P(U) \leq \delta + \varepsilon, \qquad \int_{T^{-1}(U)} \|J_T\|_{\mathrm{op}}\, d\gamma \geq \Delta I(w - \varepsilon). \tag{5.4}$$

If $\delta + \varepsilon > 0$, *then, for every* $q > 1$,

$$\begin{aligned} \big(\mathbb{E}_\gamma \|J_T\|_{\mathrm{op}}^q\big)^{1/q} &\geq \left(\int_{T^{-1}(U)} \|J_T\|_{\mathrm{op}}^q\, d\gamma\right)^{1/q} \\ &\geq \frac{\Delta I(w-\varepsilon)}{(\delta+\varepsilon)^{1-1/q}}. \end{aligned} \tag{5.5}$$

Bound (5.4) implies $\mathbb{E}_\gamma \|J_T\|_{\mathrm{op}} \geq \Delta I(w-\varepsilon)$. For fixed w and ε, this lower bound increases linearly with the separation Δ. Bound (5.5) additionally quantifies the effect of a small upper bound δ on the probability $\mu(U)$ assigned to the separating region U. The Jacobian contribution $\Delta I(w-\varepsilon)$ required by (5.4) must come from the source points mapped into U, whose total probability is at most $\delta + \varepsilon$. Concentrating this Jacobian contribution on a set of small probability forces the higher Jacobian moments to be large. In particular, for each fixed $q > 1$, bound (5.5) diverges as $\delta, \varepsilon \to 0$ if Δ and w remain bounded away from zero.

We can obtain similar results for log-concave sources, as formalized in the next proposition. For a probability measure ν with a log-concave density on $\mathbb{R}^d$, its Cheeger constant $h_\nu > 0$ is the largest constant satisfying

$$h_\nu \inf_{c \in \mathbb{R}} \mathbb{E}_\nu |g - c| \leq \mathbb{E}_\nu \|\nabla g\|_2 \tag{5.6}$$

for every bounded locally Lipschitz function g [6, 51]. The Cheeger constant and its connection to the Poincaré constant are reviewed in Section S1.6 of the Supplement.

PROPOSITION 5.3 (Multimodal bounds for log-concave sources). *Let* ν *have a log-concave density on* $\mathbb{R}^d$, *and let* $T : \mathbb{R}^d \to \mathbb{R}^m$ *be locally Lipschitz, with* $P = T_{\#}\nu$. *For* A, Δ, *and* U *as in* (5.1), *put* $b = \min\{P(A), 1 - P(A^{\Delta})\}$. *Then*

$$\int_{T^{-1}(U)} \|J_T\|_{\mathrm{op}}\, d\nu \geq h_\nu \Delta b. \tag{5.7}$$

If (5.3) *holds and* $\mathrm{TV}(P,\mu) \le \varepsilon < w = \min\{w_1, w_2\}$*, then* $P(U) \le \delta + \varepsilon$*. For* $\delta + \varepsilon > 0$ *and every* $q > 1$*,*

$$\begin{aligned}\left(\mathbb{E}_\nu \|J_T\|_{\mathrm{op}}^q\right)^{1/q} &\ge \left(\int_{T^{-1}(U)} \|J_T\|_{\mathrm{op}}^q \, d\nu\right)^{1/q} \\ &\ge \frac{h_\nu \Delta(w-\varepsilon)}{(\delta+\varepsilon)^{1-1/q}}.\end{aligned} \tag{5.8}$$

All integrals are understood in the extended nonnegative sense.

Proposition 5.3 gives a similar conclusion to Corollary 5.2 for log-concave sources. For any fixed log-concave source ν and every fixed $q > 1$, the lower bound in (5.8) diverges as $\delta, \varepsilon \to 0$, provided Δ and w remain bounded away from zero.

The strength of the bounds in (5.7) and (5.8) depends on the source's Cheeger constant h_ν. For log-concave ν,

$$h_\nu \ge c\, C_{\mathrm{PI}}(\nu)^{-1/2},$$

where $C_{\mathrm{PI}}(\nu)$ is the variance Poincaré constant defined in Section 4 [51]. Thus an upper bound on $C_{\mathrm{PI}}(\nu)$ provides a lower bound on h_ν, which can be substituted into (5.7) and (5.8).

We now illustrate these bounds using a standard Gaussian source and a two-component Gaussian mixture target.

EXAMPLE 5.4 (A Two-Component Gaussian Mixture). Fix $a > 0$ and consider the Gaussian mixture

$$\mu_\sigma = \tfrac{1}{2}\mathcal{N}(-a, \sigma^2) + \tfrac{1}{2}\mathcal{N}(a, \sigma^2), \qquad 0 < \sigma \le 1. \tag{5.9}$$

We let $\sigma \downarrow 0$ with a fixed; the mixture is bimodal for $\sigma < a$. Take $A = (-\infty, -a/2]$ and $\Delta = a$, so that $U = (-a/2, a/2)$. By symmetry, both side probabilities are $w_\sigma = (1 - \delta_\sigma)/2$, where

$$\delta_\sigma = \mu_\sigma(U) = \Phi\left(\frac{3a}{2\sigma}\right) - \Phi\left(\frac{a}{2\sigma}\right) \longrightarrow 0. \tag{5.10}$$

Consequently, every locally Lipschitz exact transport from a standard Gaussian source to μ_σ satisfies

$$\left(\mathbb{E}_\gamma \|J_T\|_{\mathrm{op}}^q\right)^{1/q} \ge \frac{a I((1-\delta_\sigma)/2)}{\delta_\sigma^{1-1/q}} \longrightarrow \infty, \qquad q > 1. \tag{5.11}$$

This divergence occurs while every fixed absolute moment of the target remains bounded.

The same behavior is visible directly in the univariate monotone transport map $T_\sigma = F_\sigma^{-1} \circ \Phi$, where F_σ and f_σ are the cumulative distribution function and density of μ_σ. Since $F_\sigma(T_\sigma(z)) = \Phi(z)$,

$$T_\sigma'(z) = \frac{\varphi(z)}{f_\sigma(T_\sigma(z))}, \qquad T_\sigma'(0) = \sigma \exp\left(\frac{a^2}{2\sigma^2}\right). \tag{5.12}$$

According to (5.12), the derivative at the center diverges as $\sigma \downarrow 0$, while the target density there tends to zero. Equation (5.11) shows that this growth in Jacobian moments is *intrinsic* to the transport problem, rather than a consequence of the particular monotone construction. Details of these calculations are given in Section S3.1 of the Supplement.

6. Practical Implications for Deep Generative Modeling. Our results concern properties of *any* pushforward map T that transports a source distribution ν to a target $\mu = T_{\#}\nu$. Our heavy-tail bounds in Section 4 apply to Gaussian and, more generally, log-concave sources under weak Sobolev regularity. Meanwhile, our multimodal bounds in Section 5 require only local Lipschitz continuity. We now outline practical implications of these results for deep generative models.

Prior Lipschitz-based Analysis. Prior work studies heavy-tailed targets under global Lipschitz assumptions on the generative mapping $\widehat{T}$ [34, 66, 72]. Related analyses address disconnected or multimodal targets under global Lipschitz or bi-Lipschitz assumptions [59, 67, 70]. A global Lipschitz bound requires $\|\widehat{T}(x) - \widehat{T}(y)\|_2 \le L\|x - y\|_2$ for all x, y. This raises the question of whether allowing mappings without such a bound removes these difficulties.

Our results show that large derivatives can remain necessary without a global Lipschitz constraint. For example, with $Z \sim N(0, I_d)$, large component variances of $T(Z)$ require $\mathbb{E}\|\nabla T_j(Z)\|_2^2$ to be large for the corresponding coordinates j. This averaged bound does not determine where large derivatives occur. For the explicit heavy-tail rearrangements in Section 8, they grow rapidly in the latent tails. For multimodal targets, our bounds identify a large Jacobian contribution over the preimage of the low-probability region separating modes, even when target moments remain bounded. The need for large gradients is therefore intrinsic to the transport problem.

Implications for Expressivity versus Stability Trade-off. The requirements of large gradients are related to a practical trade-off between numerical stability and expressivity. It is known that large gradients can contribute to numerical instability in normalizing flow models [4]. For a differentiable mapping T, perturbing an input z by an error e yields

$$T(z + e) - T(z) = J_T(z)e + o(\|e\|_2), \qquad e \to 0,$$

where $J_T(z)$ is the Jacobian at z. Large Jacobian norms therefore allow small input errors in some directions to produce much larger output errors. Such errors can arise from rounding and finite-precision representations in numerical computations. Behrmann et al. [4] document related numerical failures in reconstruction and sampling, even for mathematically invertible neural networks. Large gradients are also related to numerical instability in neural ODE models [12, 23], as detailed in the following example.

EXAMPLE 6.1. In neural ODE models, the transport map $T(z) = X_1(z)$ is obtained by solving the ODE $\dot{X}_t(z) = v_t(X_t(z))$ over $t \in [0, 1]$, with initial condition $X_0(z) = z$. We use $D_z X_t(z)$ to denote the Jacobian of the flow with respect to its initial condition z, and $D_x v_t(x)$ to denote the Jacobian of the vector field with respect to its spatial argument x. Assume that $v_t(x)$ and $D_x v_t(x)$ are continuous in (t, x) and that the flow exists on $[0, 1]$. An application of the variational equation for $D_z X_t(z)$ [68] together with Grönwall's inequality yields

$$\log \|J_T(z)\|_{\mathrm{op}} \le \int_0^1 \|D_x v_t(X_t(z))\|_{\mathrm{op}}\, dt. \tag{6.1}$$

where $\|\cdot\|_{\mathrm{op}}$ denotes the operator norm. Large transport derivatives therefore require a large time integral of the vector-field's Jacobian norm. Although the logarithm in (6.1) moderates this requirement, transport derivatives can grow rapidly as the separation between target modes increases. For the two-component Gaussian mixture in Proposition 7.4, suppose that the neural ODE realizes the monotone transport map T_D. We then have

$$\begin{aligned} \mathbb{E}\left[\exp\left(2\int_0^1 \|D_x v_t(X_t(Z))\|_{\mathrm{op}}\, dt\right)\right] &\ge \mathbb{E}\big[T_D'(Z)^2\big] \\ &\ge \frac{c}{D}\exp(D^2/8), \qquad D \ge 2, \end{aligned} \tag{6.2}$$

where D is the distance between the component means, $c > 0$, and the last inequality is due to Proposition 7.4. This illustrates how large transport derivatives can be associated with large accumulated vector-field Jacobian norms along individual trajectories.

This is relevant because neural ODEs rely on numerical solvers to integrate the learned vector field. Finlay et al. [23] associate large vector-field Jacobian norms with numerical instability in fixed-step solvers and more required function evaluations in adaptive solvers.

A natural approach to mitigating numerical stability issues associated with large transport gradients in normalizing flows is to impose a uniform upper bound on the Jacobian operator norm, thereby enforcing a global Lipschitz constraint on the transport map [70]. When such bounds are imposed on both the transport map and its inverse, the resulting bi-Lipschitz constraints control gradients in both directions [70]. While these restrictions can improve numerical stability, they limit the expressivity of the map to reproduce the target's tails and modal structure. For a Gaussian and log-concaves sources, global Lipschitz constraints rule out exact representation of heavy-tailed targets [34, 66]. For multimodal targets, a global Lipschitz bound can prevent accurate representation of well-separated modes [59]. Verine et al. [70] establish related lower bounds on total variation approximation error for bi-Lipschitz normalizing flows.

Section 7 considers simple, invertible coordinate transformations when the target density can be evaluated up to a normalizing constant. Their invertibility allows us to learn the transport in the transformed coordinates and then recover samples in the original coordinates by applying the known inverse transformation. By lightening the tails or bringing separated modes closer together in the transformed coordinates, these transformations can reduce the Jacobian norms required of the learned transport map.

7. Approximating Densities via Bijective Reparametrizations. The preceding results describe limitations of learning a pushforward map in the original coordinates of the target. Several prior works have considered modifying the source distribution to be heavier-tailed to model heavy-tailed targets [34], including in diffusion models [54]. Here, we propose a different approach which retains the simple source distribution but reparametrizes the *target* geometry through a change of coordinates. By learning in the transformed coordinates, we aim to reduce the Jacobian norms required of the transport map. We consider the task of approximating a target density that can be evaluated up to a normalizing constant.

We focus on simple one-dimensional transformations that can be applied coordinate-wise to multivariate targets. These transformations simplify marginal geometry, leaving the learned generative map to model the joint distribution, including dependence between coordinates.

Let $\Psi : \mathbb{R}^m \to \mathbb{R}^m$ be a bijection such that both Ψ and Ψ^{-1} are C^1. We choose Ψ before fitting the learned map and write $\tau = \Psi_{\#}\mu$ for the transformed target. We learn a map S from the source ν to τ, and generate a sample in the original coordinates by setting $X = \Psi^{-1}(S(Z))$, where $Z \sim \nu$. The Jacobian of the resulting map $T = \Psi^{-1} \circ S$ satisfies

$$J_T(z) = J_{\Psi^{-1}}(S(z))J_S(z). \tag{7.1}$$

The lower bounds established earlier still apply to T. However, the growth required of J_T can be captured partly by the known inverse transformation, allowing S to have a smaller expected gradient norm. We illustrate this approach with a reparametrization for heavy tails and a reparametrization for multimodal densities. Proofs are given in Section S4 of the Supplement.

7.1. *Reparametrizations for Heavy Tails.* A simple example of a bijective reparameterization is the inverse hyperbolic sine function,

$$\psi_s(x) = \operatorname{arsinh}(x/s), \qquad \psi_s^{-1}(y) = s \sinh y, \qquad s > 0. \tag{7.2}$$

This transformation is approximately linear near the origin and grows logarithmically in the tails. In particular, both its inverse and its derivative $\psi_s'(x) = (s^2 + x^2)^{-1/2}$ are available in analytical form.

We now show how this reparametrization can lighten heavy tails. Related definitions of slowly varying functions and regularly varying tails are given in Section S1.3 of the Supplement.

PROPOSITION 7.1 (Tail behavior after transformation). *Suppose that* $\mathbb{P}(|X| > x) = x^{-\alpha}L(x)$ *for all sufficiently large* x*, where* $\alpha > 0$ *and* L *is slowly varying. For any fixed* $s > 0$*, the random variable* $Y = \psi_s(X)$ *satisfies*

$$\lim_{y\to\infty} \frac{-\log \mathbb{P}(|Y| > y)}{y} = \alpha. \tag{7.3}$$

Consequently, $\mathbb{E}\exp(t|Y|) < \infty$ *for* $0 < t < \alpha$*, and every polynomial moment of* Y *is finite.*

Consequently, in one dimension, the reparametrization (7.2) converts regularly varying tails into exponential tails without requiring knowledge of the tail index. We illustrate with an example concerning the Student-t family.

EXAMPLE 7.2 (Transformed Student-t distributions). Let $X \sim t_\kappa$, where $\kappa > 0$ denotes the degrees of freedom, and apply the change of coordinates $Y = \operatorname{arsinh}(X/\sqrt{\kappa})$. Write τ_κ for the distribution of Y. Substituting $x = \sqrt{\kappa}\sinh y$ into the change-of-variables formula gives

$$f_\kappa(y) = \frac{\Gamma((\kappa+1)/2)}{\sqrt{\pi}\,\Gamma(\kappa/2)}(\cosh y)^{-\kappa}, \qquad y \in \mathbb{R}. \tag{7.4}$$

Since $\cosh y \sim e^{|y|}/2$ as $|y| \to \infty$, the transformed density decays exponentially rather than polynomially. In particular, Y has finite moments of every order. This illustrates the tail and moment conclusions of Proposition 7.1.

To see the implication for transport gradients, let $S_\kappa = F_{\tau_\kappa}^{-1} \circ \Phi$, where F_{τ_κ} and Φ are the cumulative distribution functions of τ_κ and the standard Gaussian, respectively. Then $S_\kappa(Z) \sim \tau_\kappa$ for $Z \sim N(0,1)$. For $0 < \kappa \le 2$, the monotone transport map to the transformed target satisfies

$$\mathbb{E}\big[(S_\kappa'(Z))^2\big] < \infty, \tag{7.5}$$

as established in Section S4.2 of the Supplement. Thus, changing coordinates yields an exact transport with finite expected squared derivative.

More generally, the reparametrization (7.2) can be applied coordinate-wise to lighten the marginal tails of a multivariate target, without requiring independence between coordinates.

7.2. *Reparametrizations for Multimodal Target Densities.* Approximating multimodal target densities without additional structural assumptions is challenging, particularly when the locations of their modes are unknown [55, 59]. We therefore focus on a setting in which reference component locations are available. These locations may be supplied through prior knowledge of the model structure or preliminary location estimates. For example, label permutations in Bayesian mixture models [65] and sign symmetries in Bayesian factor analysis [20] allow an approximate estimate of one mode to generate reference locations for symmetry-related modes. We use these reference locations to construct a bijective reparametrization that

brings separated components closer together in the transformed coordinates, which can reduce the Jacobian norms required of the learned transport map.

To illustrate this in one dimension, let $c_1 < \cdots < c_K$ be the reference component locations. Choose $r, h > 0$. The parameter r sets the radius of neighborhoods around these locations whose widths we preserve, while h controls the compression between them. Let $d(x) = \min_i |x - c_i|$ denote the distance to the nearest reference location. We restrict compression to the interval $[L_\Psi, U_\Psi]$, where $L_\Psi = c_1 - r$ and $U_\Psi = c_K + r$.

We define the change of coordinates by

$$\Psi(x) = \int_0^x a(t)\, dt, \tag{7.6}$$

where its derivative $\Psi'(x) = a(x)$ is given by

$$a(x) = \begin{cases} \exp\left[-\dfrac{(d(x) - r)_+^2}{2h^2}\right], & L_\Psi \le x \le U_\Psi, \\ 1, & x < L_\Psi \text{ or } x > U_\Psi, \end{cases} \tag{7.7}$$

and $(u)_+ = \max\{u, 0\}$. Within each neighborhood $[c_i - r, c_i + r]$, we have $a(x) = 1$, so Ψ simply translates the interval without changing its width. Between these neighborhoods, $a(x) < 1$, so Ψ compresses the gaps and brings the components closer together. Smaller h produces stronger compression.

The derivative (7.7) remains strictly positive everywhere and equals one outside $[L_\Psi, U_\Psi]$, ensuring that Ψ is invertible on $\mathbb{R}$. The construction of Ψ uses only the reference component locations and the parameters r and h, without requiring knowledge of the target CDF or component weights.

We illustrate this reparametrization with a two-component Gaussian mixture, comparing transport gradients in the original and transformed coordinates. Consider the model,

$$\mu_D = \tfrac{1}{2}N(-D/2, 1) + \tfrac{1}{2}N(D/2, 1), \qquad D > 0,$$

where D is the separation between the component means. Write F_D and Φ for the cumulative distribution functions of μ_D and the standard Gaussian, respectively. The monotone transport map from $N(0,1)$ to μ_D is $T_D = F_D^{-1} \circ \Phi$.

PROPOSITION 7.3 (Derivative bound after reparametrization). *Fix $r > 0$ and $0 < h < 1$. Assume that $0 \le \eta \le r$. Suppose the reference component locations $c_1 < c_2$ approximate the component means with error at most η:*

$$|c_1 + D/2| \le \eta, \qquad |c_2 - D/2| \le \eta.$$

Construct Ψ using (7.6)–(7.7), and set $S_D = \Psi \circ T_D$. Then S_D transports $N(0,1)$ to the transformed target $\Psi_\# \mu_D$, and

$$0 < S_D'(z) \le C, \qquad C = 2\exp\left(\frac{(r+\eta)^2}{2(1-h^2)}\right), \qquad z \in \mathbb{R}. \tag{7.8}$$

Consequently, $\mathbb{E}[S_D'(Z)^2] \le C^2$ for $Z \sim N(0,1)$, with C independent of D.

The quantity r reflects the assumed accuracy of the reference component locations, as a smaller assumed error η allows a smaller r while maintaining $\eta \le r$. The assumption $\eta \le r$ ensures that each component mean lies within the neighborhood around its reference location so that the transformed transport can have uniformly bounded gradients even as the original component separation increases. We next quantify the gradient growth of T_D in the original coordinates.

PROPOSITION 7.4 (Growth of the original transport). *For the same mixture with $D \geq 2$, there is an absolute constant $c > 0$, independent of D, such that, for $Z \sim N(0,1)$,*

$$\mathbb{E}[T_D'(Z)^2] \geq \frac{c}{D} e^{D^2/8}. \tag{7.9}$$

The lower bound in (7.9) grows exponentially with the squared component separation. Increasing separation therefore forces rapid growth in the expected squared derivative of the original transport. Proofs of Propositions 7.3–7.4 are given in Section S4.3 of the Supplement.

8. Empirical Illustrations. We use explicit transport maps from standard Gaussian sources to examine how target geometry affects transport gradients. Sections 8.1 and 8.2 illustrate the gradient growth associated with heavy tails and separated modes. Section 8.3 then examines how the reparametrizations from Section 7 affect these gradients. Using known transport maps allows us to study these effects without approximation or optimization error. We provide additional empirical illustrations in Section S5 of the Supplement.

8.1. *Heavy-Tailed Targets.* We provide empirical illustrations of our theoretical results through numerical simulations on commonly used families of heavy-tailed target distributions. Specifically, we consider the Student-t family with varying degrees of freedom κ, as well as the Pareto and Fréchet families with different shape parameters α and ζ respectively. We review related definitions of these canonical families in Section S1.3 of the Supplement. As κ, α, ζ decrease, the tails of the corresponding family become heavier.

In one dimension, an explicit transport map between probability measures can be constructed using quantile functions. Let ν and μ be probability measures on $\mathbb{R}$ with continuous cumulative distribution functions F_ν and F_μ. If $Z \sim \nu$, then the map,

$$T(z) = F_\mu^{-1}\big(F_\nu(z)\big), \tag{8.1}$$

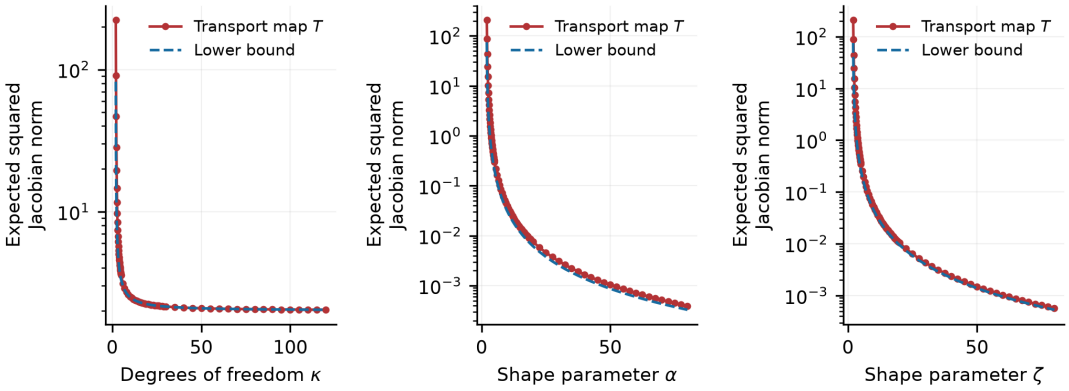
pushes ν forward to μ, i.e., $T_\# \nu = \mu$. This monotone transport map is also known as the monotone rearrangement [71] and enjoys optimality properties [60].

For each target family, we evaluate the transport map in (8.1) from a standard Gaussian source. Figure 2 compares the numerically evaluated expected squared Frobenius norm of its Jacobian with the Poincaré lower bound as the target's tail parameter varies. Figure 3 depicts the log absolute Jacobian determinant across the source space. Section S5.1 of the Supplement provides further implementation details.

Figure 2 shows that the Poincaré lower bounds from Theorem 4.2 closely track the numerically evaluated expected squared Frobenius norms across the three target families. Both quantities increase sharply as the tail parameters κ, α, ζ approach 2 from above, where the corresponding target variances diverge. These comparisons illustrate how the moment-based bounds capture the growth of transport gradients as the targets become heavier-tailed.

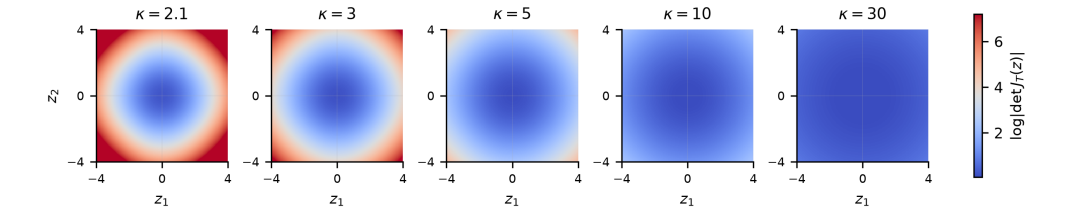
Figure 3 depicts where the transport expands the source distribution to populate the target's heavy tails. The log absolute Jacobian determinant increases in these regions as the tails become heavier. The Student-t targets have heavy tails in both directions, whereas the Pareto and Fréchet targets are right-skewed and have a single heavy tail.

8.2. *Multimodal Targets.* We next consider Gaussian mixture targets to illustrate the bounds in Section 5. As in the heavy-tailed examples, the monotone transport map allows us to examine the transport directly, without introducing approximation or optimization error. We use the one-dimensional mixture $\mu = \frac{1}{2}\mathcal{N}(-a, \sigma^2) + \frac{1}{2}\mathcal{N}(a, \sigma^2)$, where $a = \Delta/2 + 3\sigma$, and take $U = (-\Delta/2, \Delta/2)$ as the valley. Increasing Δ separates the components while keeping their scales fixed.

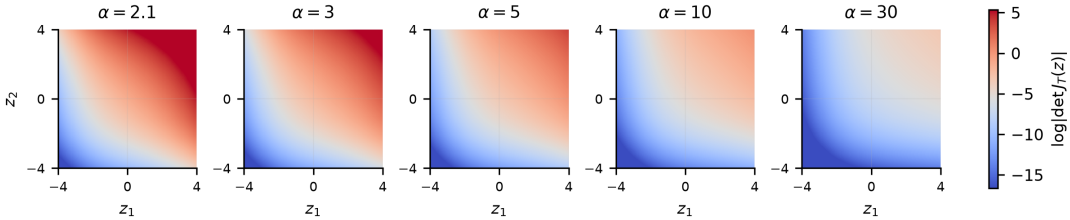


(a) Student-t target (b) Pareto target (c) Fréchet target

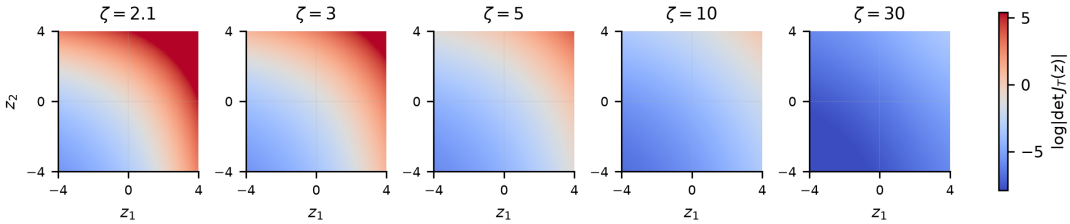
Fig 2: Comparison of Poincaré lower bounds with the expected squared Frobenius norm of the transport Jacobian for different heavy-tailed target distributions under a standard Gaussian source. The red curves with circular markers show the numerically evaluated quantities $\mathbb{E}[\|J_T(Z)\|_F^2]$. The blue dashed curves show the lower bounds.



(a) Student-t target



(b) Pareto target



(c) Fréchet target

Fig 3: Log absolute Jacobian determinant heatmaps for Gaussian-to-heavy-tailed transport maps. Note that the Pareto and Fréchet distributions are right-skewed and supported on the positive orthant.

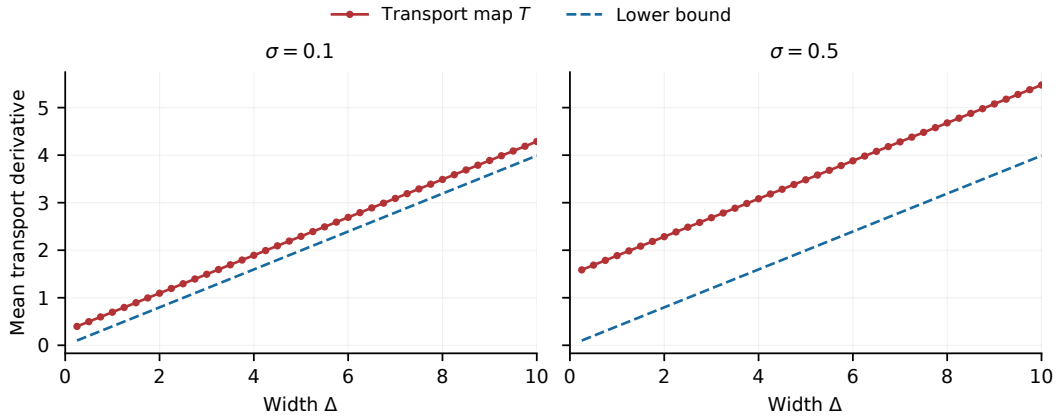


Fig 4: Gradient bounds for two-component Gaussian mixtures. The plots compare $\mathbb{E}[T'(Z)]$ with the lower bound in Theorem 5.1 as the width Δ of the low-probability interval increases. The two panels use component standard deviations $\sigma = 0.1$ and $\sigma = 0.5$.

Figure 4 compares the numerically evaluated mean derivative $\mathbb{E}[T'(Z)]$ with the lower bound in Theorem 5.1. Here, T is the monotone transport map in (8.1) from the standard Gaussian source to the mixture target. The expectation averages its derivative over source points $Z \sim N(0,1)$. Both the lower bound and the computed mean derivative increase with the width of the low-probability interval. We observe that the lower bound captures the rate of this increase well.

8.3. *Reparametrizations of Known Transport Maps.* We next examine how changing the target coordinates affects transport derivatives. For each scenario below, we consider a known map T that transports a standard Gaussian source to the original target, so $X = T(Z)$ is a sample from that target. Applying the fixed coordinate transformation Ψ gives $Y = \Psi(X)$. The composition $S = \Psi \circ T$ therefore transports the same Gaussian source to the target expressed in transformed coordinates. We compare $\mathbb{E}[\|J_T(Z)\|_F^2]$ and $\mathbb{E}[\|J_S(Z)\|_F^2]$, which measure the squared Jacobian norms averaged over the Gaussian source. In one dimension, these quantities are $\mathbb{E}[T'(Z)^2]$ and $\mathbb{E}[S'(Z)^2]$, respectively.

For the heavy-tailed examples, we use the three canonical families from Section 8.1, namely the Student-t, Pareto, and Fréchet distributions. Each target is a two-dimensional product distribution with independent coordinates, and the source is $N(0, I_2)$. We apply the asinh transformation in (7.2) coordinate-wise, using $s = \sqrt{\kappa}$ for the Student-t family and $s = 1$ for the Pareto and Fréchet families. We vary each family's tail parameter from 1 to 10.

For the multimodal examples, we use the one-dimensional Gaussian mixture

$$\mu_D = \frac{1}{5}\sum_{i=1}^{5} N(m_i, 1), \qquad m_i = D(i-3), \qquad D \in \{3, 4, \ldots, 16\}.$$

Thus, D controls the separation between adjacent component means while the component weights and variances remain fixed. We construct Ψ using (7.6)–(7.7), with $r = 1.5$ and $h = 0.4$. We specify approximate reference component locations c_i satisfying $|c_i - m_i| \le 0.5$. The differences $c_i - m_i$ are listed in Section S5.4 of the Supplement, which also describes the numerical integration procedure.

Figure 5 shows that the asinh reparametrization substantially reduces the expected squared Jacobian norms across all three target families. The original-map values increase sharply as

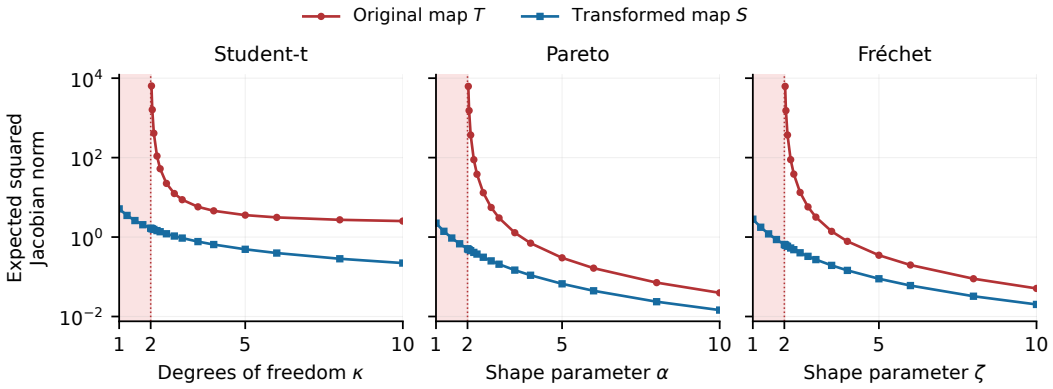


Fig 5: Expected squared Frobenius Jacobian norms before and after the asinh reparametrization for two-dimensional product Student-t, Pareto, and Fréchet targets, using a standard Gaussian source. The shaded regions indicate degrees of freedom or shape parameters less than or equal to 2.

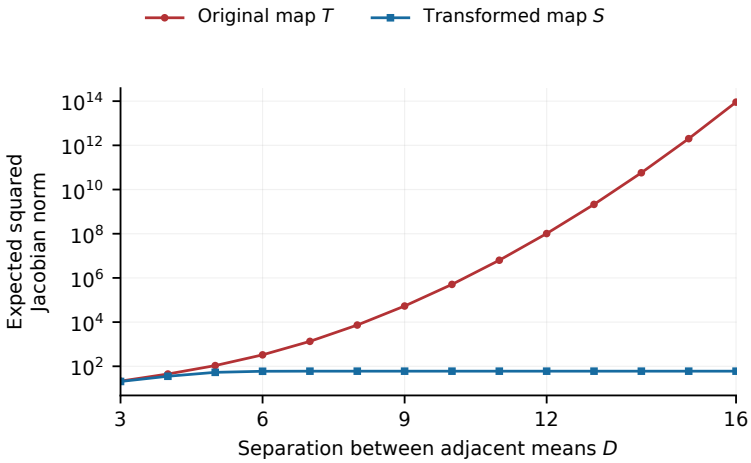


Fig 6: Expected squared derivatives $\mathbb{E}[T'(Z)^2]$ and $\mathbb{E}[S'(Z)^2]$, for $Z \sim N(0,1)$, before and after transformation for five-component Gaussian mixtures as adjacent means move farther apart. Component widths, transformation parameters, and errors in the reference component locations are held fixed.

the tail parameter approaches 2 from above. The transformed-map values remain moderate across the displayed range. These results are in line with Proposition 7.1, which explains how the transformation converts heavy tails into exponential tails.

Figure 6 shows that the original transport's expected squared derivative increases rapidly as adjacent component means move farther apart. After reparametrization, the corresponding quantity levels off, although the component separation continues to increase. This five-component example exhibits the same qualitative contrast established for two-component mixtures by Propositions 7.3 and 7.4: the transformed transport admits a derivative bound independent of separation, whereas the original transport's expected squared derivative grows exponentially with squared separation.

Declaration of AI Use. The authors developed all core logic and theoretical frameworks in this paper and drafted the initial manuscript. AI tools (ChatGPT 6 Astra and Claude Opus 5.5) were used to refine the writing, check mathematical derivations, and assist in code writing. After using these tools, the authors carefully reviewed and edited the content as necessary. The authors take full responsibility for the final manuscript.

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Supplement to "Intrinsic Gradient Barriers in Pushforward Deep Generative Models".

S1. Technical Preliminaries.

S1.1. *Weak Gradients.* Let ν be a full-dimensional log-concave probability measure on $\mathbb{R}^d$, and set $\Omega = \operatorname{int}(\operatorname{supp}\nu)$. A function $f \in L^p(\nu)$ that is locally integrable on Ω is said to admit a *weak gradient* if there exists a vector-valued function $g = (g_1, \dots, g_d) \in L^p(\nu; \mathbb{R}^d)$, also locally integrable on Ω, such that

$$\int_\Omega f(x)\, \partial_i \varphi(x)\, dx \;=\; -\int_\Omega g_i(x)\, \varphi(x)\, dx,$$

for all smooth, compactly supported test functions $\varphi \in C_c^\infty(\Omega)$ and all $i = 1, \dots, d$. In this case, we write $\nabla f = g$ and refer to ∇f as the weak gradient of f.

Examples.. For functions that are classically differentiable, the weak gradient coincides with the classical gradient, so the definition introduces nothing new for smooth maps. The notion becomes useful for functions with isolated non-smoothness:

- The function $f(x) = |x|$ on $\mathbb{R}$ is not differentiable at 0, but admits a weak derivative $f'(x) = \operatorname{sgn}(x)$, defined almost everywhere.
- The ReLU function $\operatorname{ReLU}(x) = \max(x, 0)$ admits the weak derivative $\mathbf{1}\{x > 0\}$.

Generally, on an intuitive level, piecewise-smooth functions with finitely many "kinks" admit weak gradients given by the classical gradient on each smooth piece. This level of generality is the natural one in our setting for three reasons: (i) many neural-network-based maps are only piecewise smooth; (ii) weak derivatives are the standard notion of derivative in Sobolev spaces and in the formulation of Poincaré-type inequalities; and (iii) they are the natural notion of derivative used in PDE-based analyses of flow- and transport-based generative models. We refer the reader to standard texts on Sobolev spaces and weak derivatives, such as [21], for further background.

S1.2. *Sobolev Spaces.* Any rigorous analysis of pushforward maps requires minimal regularity assumptions to ensure that gradients and expectations thereof are well defined. Since our main analytical tools are Poincaré inequalities, which relate moments of a distribution to integrals involving gradients, we work with weak notions of differentiability formalized through Sobolev spaces.

For $1 \le p < \infty$, $L^p(\nu)$ is the space of measurable functions f satisfying $\int |f|^p\, d\nu < \infty$, with norm

$$\|f\|_{L^p(\nu)} = \left(\int |f(x)|^p\, d\nu(x) \right)^{1/p}.$$

For $p = \infty$, the norm is the essential supremum with respect to ν.

For $1 \le p < \infty$, the Sobolev space $W^{1,p}(\nu)$ consists of all functions $f \in L^p(\nu)$ that admit a weak gradient $\nabla f \in L^p(\nu; \mathbb{R}^d)$. The associated norm is defined by

$$\|f\|^p_{W^{1,p}(\nu)} \;=\; \|f\|^p_{L^p(\nu)} + \|\nabla f\|^p_{L^p(\nu)}.$$

In particular, the space $W^{1,2}(\nu)$ plays a central role in our analysis, as it provides the natural domain for Poincaré and log-Sobolev inequalities. Membership in $W^{1,2}(\nu)$ requires only weak differentiability and square-integrability of the gradient, imposing substantially weaker regularity than global Lipschitz conditions. We refer to [21] and [10] for comprehensive treatments of Sobolev spaces and weak derivatives, and [7] for details on Gaussian Sobolev spaces.

S1.3. *Background on Heavy-Tailed Distributions.* A central object in this work is the tail behavior of the target distribution. Tail decay is closely tied to the existence and growth of moments. For a real-valued random variable X, we use the two-sided convention: X is heavy-tailed if $\mathbb{E}[e^{t|X|}] = \infty$ for every $t > 0$. In practice, heavy-tailed distributions typically possess only finitely many moments, whereas light-tailed distributions admit moments of all orders. We briefly review relationships between heavy tails, p-th moments, and polynomial tail bounds; see [57] for additional background.

Moments and Polynomial Tail Bounds. Let $X \sim \nu$. For $p \geq 1$, the p-th moment of ν is

$$\mathbb{E}_\nu \|X\|^p \;=\; \int_{\mathbb{R}^d} \|x\|^p \, d\nu(x),$$

whenever the integral is finite. A basic consequence of Markov's inequality is the polynomial tail bound

$$\mathbb{P}\big(\|X\| \geq t\big) \;\leq\; \frac{\mathbb{E}\|X\|^p}{t^p}, \qquad t > 0, \tag{S1.1}$$

which shows that finiteness of the p-th moment implies at least t^{-p} control of the tail probability.

Conversely, moments can be expressed in terms of the tail function via the standard identity

$$\mathbb{E}\|X\|^p \;=\; p \int_0^\infty t^{p-1}\, \mathbb{P}\big(\|X\| \geq t\big)\, dt, \tag{S1.2}$$

whenever the integral is finite. In particular, if $\mathbb{P}(\|X\| \geq t) \lesssim t^{-\alpha}$ for some $\alpha > 0$, then (S1.2) implies $\mathbb{E}\|X\|^p < \infty$ for all $p < \alpha$. Conversely, if $\mathbb{P}(\|X\| \geq t) \gtrsim t^{-\alpha}$ for large t, then moments of order $p \geq \alpha$ diverge. Thus, polynomial tail decay corresponds to the existence of moments only up to a critical order.

Complementary lower bounds on tail probabilities can be obtained under additional assumptions via the Paley–Zygmund inequality. Specifically, if $Y \geq 0$ has finite second moment, then for any $0 < s < 1$,

$$\mathbb{P}\big(Y \geq s\,\mathbb{E}Y\big) \;\geq\; (1-s)^2\, \frac{(\mathbb{E}Y)^2}{\mathbb{E}Y^2}.$$

Applying this inequality with $Y = \|X\|^p$ yields

$$\mathbb{P}\Big(\|X\| \geq s^{1/p} (\mathbb{E}\|X\|^p)^{1/p}\Big) \;\geq\; (1-s)^2\, \frac{(\mathbb{E}\|X\|^p)^2}{\mathbb{E}\|X\|^{2p}},$$

which shows that, provided $\mathbb{E}\|X\|^{2p}$ is controlled relative to $(\mathbb{E}\|X\|^p)^2$, large p-th moments necessarily imply non-negligible mass in the tails. We refer to [18] for further discussion.

Slow and Regular Variation. We recall the following standard definitions; see Bingham et al. [5] for further background.

A positive measurable function L is *slowly varying at infinity* if

$$\lim_{x \to \infty} \frac{L(cx)}{L(x)} = 1 \qquad \text{for every fixed } c > 0.$$

A positive measurable function g is *regularly varying at infinity with index* $\rho \in \mathbb{R}$ if

$$\lim_{x \to \infty} \frac{g(cx)}{g(x)} = c^\rho \qquad \text{for every fixed } c > 0.$$

Equivalently, $g(x) = x^\rho L(x)$ for a slowly varying function L. Slow variation corresponds therefore to the special case $\rho = 0$.

We say that a real-valued random variable X has *regularly varying tails with tail index* $\alpha > 0$ if its tail function $x \mapsto \mathbb{P}(|X| > x)$ is regularly varying with index $-\alpha$. Equivalently,

$$\mathbb{P}(|X| > x) = x^{-\alpha} L(x)$$

for all sufficiently large x, where L is slowly varying. This describes power-law tail decay with a slowly varying multiplicative factor. For example, a Student-t distribution with κ degrees of freedom has tail index κ.

Some Canonical Heavy-Tailed Distributions. In our simulations, we consider three classical families of distributions that exhibit heavy-tailed behavior. We give more details on these families below. The Student-t distribution is a classical family of heavy-tailed distributions that interpolates between Gaussian and power-law behavior. A univariate Student-t random variable with $\kappa > 0$ degrees of freedom has density,

$$f(z) = \frac{\Gamma\left(\frac{\kappa+1}{2}\right)}{\sqrt{\kappa\pi}\,\Gamma\left(\frac{\kappa}{2}\right)} \left(1 + \frac{z^2}{\kappa}\right)^{-(\kappa+1)/2}. \tag{S1.3}$$

The degrees of freedom parameter κ controls the tail behavior of the Student-t distribution. As $\kappa \to \infty$, the Student-t converges to a standard Gaussian. As κ becomes smaller, it exhibits increasingly heavy-tailed behavior.

Unlike the Student-t family, the Pareto distribution exhibits an exact polynomially decaying right tail. A univariate Pareto random variable with shape parameter $\alpha > 0$ and scale parameter $\lambda > 0$ has density,

$$f(z) = \alpha\, \lambda^\alpha\, z^{-(\alpha+1)},\ z \geq \lambda. \tag{S1.4}$$

The Fréchet distribution is another classical heavy-tailed distribution with a polynomially decaying right tail. A univariate Fréchet random variable with shape parameter $\zeta > 0$ and scale parameter $\lambda > 0$ has density,

$$f(z) = \zeta\, \lambda^\zeta\, z^{-(\zeta+1)} \exp\left(-\left(\frac{\lambda}{z}\right)^\zeta\right),\ z > 0. \tag{S1.5}$$

In both the Pareto and Fréchet families, the respective shape parameters α and ζ directly control tail heaviness and moment existence, in the sense that the kth moment exists if and only if $k < \alpha$ and $k < \zeta$ respectively. The smaller α and ζ are, the heavier the upper tails. In particular, the variances for the Pareto and Fréchet are finite only for $\alpha, \zeta > 2$, and they diverge as $\alpha, \zeta \downarrow 2$.

S1.4. *Log-Concave and Strongly Log-Concave Distributions.* A distribution with density p on $\mathbb{R}^d$ is *log-concave* if $\log p$ is concave, with the convention $\log 0 = -\infty$. Equivalently, $p(x) \propto e^{-V(x)}$ for a convex function V, allowing $V = +\infty$ where $p = 0$. In terms of the density, this means that

$$p((1-\theta)x + \theta y) \geq p(x)^{1-\theta} p(y)^\theta, \qquad x, y \in \mathbb{R}^d, \quad 0 < \theta < 1.$$

Thus, the density cannot dip below the geometric interpolation of its values at two points. Gaussian and Laplace distributions, and uniform distributions on convex bodies, are examples.

The distribution is *t-strongly log-concave*, for $t > 0$, if $V(x) - t\|x\|_2^2/2$ is convex. When $V \in C^2(\mathbb{R}^d)$, this is equivalent to

$$\nabla^2 V(x) \succeq t I_d, \qquad x \in \mathbb{R}^d.$$

This condition places a positive lower bound on the curvature of V in every unit direction. The standard Gaussian is 1-strongly log-concave. See Saumard and Wellner [61] for a detailed review, and Chewi [14] and Bakry et al. [2] for related functional inequalities.

S1.5. *Isoperimetric Profiles.* An isoperimetric profile describes the smallest boundary measure of a region having a specified probability. For a Borel set $A \subseteq \mathbb{R}^d$, let $A^\varepsilon = \{x : \operatorname{dist}(x, A) \le \varepsilon\}$. Its boundary measure under ν is

$$\nu^+(A) = \liminf_{\varepsilon \downarrow 0} \frac{\nu(A^\varepsilon) - \nu(A)}{\varepsilon}.$$

This measures how quickly probability is gained when the set is slightly enlarged. The isoperimetric profile of ν is

$$I_\nu(s) = \inf_{\substack{A \subseteq \mathbb{R}^d \text{ Borel} \\ \nu(A) = s}} \nu^+(A), \qquad 0 < s < 1.$$

Thus, every region of probability s has boundary measure at least $I_\nu(s)$.

For the standard Gaussian measure γ in any dimension,

$$I_\gamma(s) = I(s) = \varphi(\Phi^{-1}(s)), \qquad 0 < s < 1,$$

where φ and Φ are the standard univariate Gaussian density and CDF. Half-spaces attain this bound. This gives the interpretation used in Section 5: separating regions of prescribed Gaussian probability requires a boundary of at least this size. See Ledoux [42], Milman [51], and Chewi [14] for further background.

S1.6. *Cheeger and Poincaré Constants.* Using the boundary measure defined above, the *Cheeger constant* of ν is

$$h_\nu = \inf_{\substack{A \subseteq \mathbb{R}^d \text{ Borel} \\ 0 < \nu(A) < 1}} \frac{\nu^+(A)}{\min\{\nu(A), 1 - \nu(A)\}}.$$

A larger h_ν means that separating two regions with substantial probability requires a larger boundary measure. Equivalently, h_ν is the largest constant satisfying

$$h_\nu \inf_{c \in \mathbb{R}} \mathbb{E}_\nu |g - c| \le \mathbb{E}_\nu \|\nabla g\|_2$$

for every bounded locally Lipschitz function g. This is the form used in Proposition 5.3.

The *variance Poincaré constant* $C_{\mathrm{PI}}(\nu)$ is the smallest constant satisfying

$$\operatorname{Var}_\nu(f) \le C_{\mathrm{PI}}(\nu)\, \mathbb{E}_\nu \|\nabla f\|_2^2, \qquad f \in W^{1,2}(\nu).$$

Smaller values give stronger control of a function's variance through its expected squared gradient. For the standard Gaussian measure, $C_{\mathrm{PI}}(\gamma) = 1$.

For log-concave ν, the two constants are related by

$$h_\nu \ge c\, C_{\mathrm{PI}}(\nu)^{-1/2}$$

for a universal constant $c > 0$. This connects the source-dependent constants in our moment-based and multimodal gradient bounds. See Ledoux [42], Bakry et al. [2], and Milman [51] for further background.

S2. Proofs for Section 4.

PROOF OF LEMMA 4.1. This result and proof can be found in Theorem 1.6.4 of [7]. □

PROOF OF THEOREM 4.2. (i) Apply the scalar Gaussian Poincaré inequality to each component $T_j \in W^{1,2}(\gamma)$.

(ii) Fix $a \in \mathbb{R}^m$ and consider $g_a(z) = a^\top T(z)$. Since $g_a \in W^{1,2}(\gamma)$ and $\nabla g_a(z) = J_T(z)^\top a$, the scalar Poincaré inequality yields

$$\begin{aligned}\mathrm{Var}(a^\top T(Z)) &\le \mathbb{E}\big[\|J_T(Z)^\top a\|_2^2\big] \\ &= a^\top \mathbb{E}\big[J_T(Z) J_T(Z)^\top\big] a.\end{aligned}$$

Using $\mathrm{Var}(a^\top T(Z)) = a^\top \mathrm{Cov}(T(Z)) a$ for all a gives $\mathrm{Cov}(T(Z)) \preceq \mathbb{E}[J_T(Z) J_T(Z)^\top]$.

(iii) Taking traces of the PSD inequality in (ii) gives

$$\begin{aligned}\mathrm{Tr}[\mathrm{Cov}(T(Z))] &\le \mathrm{Tr}\big(\mathbb{E}[J_T(Z) J_T(Z)^\top]\big) \\ &= \mathbb{E}\ \mathrm{Tr}(J_T(Z) J_T(Z)^\top) = \mathbb{E}\|J_T(Z)\|_F^2.\end{aligned}$$

(iv) Let $f = T_i$ and $g = T_j$. By Cauchy–Schwarz,

$$\begin{aligned}&|\mathrm{Cov}(f(Z), g(Z))| \\ &\quad = \big|\mathbb{E}[(f(Z) - \mathbb{E}f(Z))(g(Z) - \mathbb{E}g(Z))]\big| \\ &\quad \le \sqrt{\mathrm{Var}(f(Z))}\,\sqrt{\mathrm{Var}(g(Z))}.\end{aligned}$$

Applying the scalar Gaussian Poincaré inequality to f and g gives $\mathrm{Var}(f(Z)) \le \mathbb{E}\|\nabla f(Z)\|_2^2$ and $\mathrm{Var}(g(Z)) \le \mathbb{E}\|\nabla g(Z)\|_2^2$, hence

$$|\mathrm{Cov}(f(Z), g(Z))| \le \sqrt{\mathbb{E}\|\nabla f(Z)\|_2^2}\,\sqrt{\mathbb{E}\|\nabla g(Z)\|_2^2}.$$

□

PROOF OF LEMMA 4.3. This result and proof can be found in Corollary 1.7.3 of [7]. □

PROOF OF COROLLARY 4.4. Apply Lemma 4.3 to each component T_j:

$$\|T_j - \mathbb{E}_\gamma T_j\|_{L^p(\gamma)} \le C_p\, \big\|\|\nabla T_j\|_2\big\|_{L^p(\gamma)}.$$

Raising to the power p and summing over j gives

$$\sum_{j=1}^m \|T_j - \mathbb{E}_\gamma T_j\|_{L^p(\gamma)}^p \le C_p^p \sum_{j=1}^m \mathbb{E}_\gamma \|\nabla T_j(Z)\|_2^p.$$

Next, since $\|J_T(z)\|_F^2 = \sum_{j=1}^m \|\nabla T_j(z)\|_2^2$, for $p \ge 2$ we have pointwise

$$\begin{aligned}\sum_{j=1}^m \|\nabla T_j(z)\|_2^p &= \sum_{j=1}^m \big(\|\nabla T_j(z)\|_2^2\big)^{p/2} \\ &\le \Big(\sum_{j=1}^m \|\nabla T_j(z)\|_2^2\Big)^{p/2} \\ &= \|J_T(z)\|_F^p.\end{aligned}$$

Taking expectations yields $\sum_{j=1}^m \mathbb{E}_\gamma \|\nabla T_j(Z)\|_2^p \le \mathbb{E}_\gamma \|J_T(Z)\|_F^p$. This completes the proof with $C_p' := C_p^p$. □

PROOF OF THEOREM 4.5. Suppose first that $d = 1$. Since ν is isotropic, $\mathrm{Var}_\nu(x) = 1$. Corollary 4.3 of [6] gives

$$C_{\mathrm{PI}}(\nu) \le 12\mathrm{Var}_\nu(x) = 12.$$

Here we use that the spectral gap in that reference is the reciprocal of our variance Poincaré constant.

For $d \ge 2$, Theorem 1.2 together with display (1.4) of [39] yields

$$C_{\mathrm{PI}}(\nu) \le C_0 \log d$$

for a universal constant $C_0 > 0$. Consequently, in every dimension $d \ge 1$,

$$C_{\mathrm{PI}}(\nu) \le C \log(ed), \qquad C = \max\{12, C_0\}.$$

The corresponding Poincaré inequalities apply to the smooth functions $f \in W^{1,2}(\nu)$ in the theorem, which proves the claim. □

PROOF OF THEOREM 4.6. Theorem 4.5 establishes the usual Poincaré inequality for ν. By Theorem 2.4 and Lemma 2.1 of [51], for every $2 \le p < \infty$,

$$\|f - \mathrm{med}_\nu(f)\|_{L^p(\nu)} \le \widetilde{C}_{p,d,\nu} \big\| \|\nabla f\|_2 \big\|_{L^p(\nu)},$$

where $\mathrm{med}_\nu(f)$ is a median of f under ν, and $\widetilde{C}_{p,d,\nu}$ depends on p and the Poincaré constant of ν. Moreover,

$$\|f - \mathbb{E}_\nu f\|_{L^p(\nu)} \le \|f - \mathrm{med}_\nu(f)\|_{L^p(\nu)} + |\mathbb{E}_\nu f - \mathrm{med}_\nu(f)| \le 2\|f - \mathrm{med}_\nu(f)\|_{L^p(\nu)}.$$

Combining these inequalities proves the result with $C_{p,d,\nu} = 2\widetilde{C}_{p,d,\nu}$. □

REMARK S2.1. The Poincaré inequalities above extend from smooth functions to their respective Sobolev classes with the same constants. These classes include locally Lipschitz functions with the required function and gradient integrability.

Proposition 27 of Barthe and Klartag [3] establishes that restrictions of functions in $C_c^\infty(\mathbb{R}^d)$ to Ω are dense in $W^{1,2}(\nu)$, with weak derivatives taken on Ω. Their proof applies equally with L^p norms replacing L^2 norms, giving density in $W^{1,p}(\nu)$ for every $1 \le p < \infty$.

Thus, for $f \in W^{1,p}(\nu)$, choose smooth $f_n \to f$ in the Sobolev norm. We then have

$$\|(f_n - \mathbb{E}_\nu f_n) - (f - \mathbb{E}_\nu f)\|_{L^p(\nu)} \le 2\|f_n - f\|_{L^p(\nu)} \longrightarrow 0.$$

Convergence in the Sobolev norm also gives $\|\nabla f_n\|_{L^p(\nu)} \to \|\nabla f\|_{L^p(\nu)}$. Taking limits in the Poincaré inequality for f_n therefore proves the same inequality for f, with the same constant. See Barthe and Klartag [3, Proposition 27 and its proof] for details of this type of standard density and closure argument.

PROOF OF PROPOSITION 4.7. Let $X \sim \mu$ and $Y \sim \widehat{\mu}$ be any coupling. Recall that

$$\mathrm{Tr}[\mathrm{Cov}(\mu)] = \mathbb{E}\|X - \mathbb{E}X\|_2^2, \qquad \mathrm{Tr}[\mathrm{Cov}(\widehat{\mu})] = \mathbb{E}\|Y - \mathbb{E}Y\|_2^2.$$

We can decompose the centered target vector as

$$X - \mathbb{E}X = (Y - \mathbb{E}Y) + \big((X - Y) - \mathbb{E}(X - Y)\big).$$

The triangle inequality in L^2 therefore gives

$$\begin{aligned}\sqrt{\mathrm{Tr}[\mathrm{Cov}(\mu)]} &= \big(\mathbb{E}\|X - \mathbb{E}X\|_2^2\big)^{1/2} \\ &\le \big(\mathbb{E}\|Y - \mathbb{E}Y\|_2^2\big)^{1/2} + \big(\mathbb{E}\|(X - Y) - \mathbb{E}(X - Y)\|_2^2\big)^{1/2}.\end{aligned}$$

Moreover, we center $X - Y$ by subtracting its mean. Expanding the squared Euclidean norm gives

$$\begin{aligned}\mathbb{E}\|(X-Y)-\mathbb{E}(X-Y)\|_2^2 &= \mathbb{E}\|X-Y\|_2^2 \\ &\quad - 2\,\mathbb{E}\langle X-Y, \mathbb{E}(X-Y)\rangle \\ &\quad + \|\mathbb{E}(X-Y)\|_2^2.\end{aligned}$$

Since $\mathbb{E}(X-Y)$ is a fixed vector, linearity of expectation yields

$$\begin{aligned}\mathbb{E}\langle X-Y, \mathbb{E}(X-Y)\rangle &= \langle \mathbb{E}(X-Y), \mathbb{E}(X-Y)\rangle \\ &= \|\mathbb{E}(X-Y)\|_2^2.\end{aligned}$$

Substituting this identity into the expansion and combining terms gives

$$\begin{aligned}\mathbb{E}\|(X-Y)-\mathbb{E}(X-Y)\|_2^2 &= \mathbb{E}\|X-Y\|_2^2 - \|\mathbb{E}(X-Y)\|_2^2 \\ &\le \mathbb{E}\|X-Y\|_2^2.\end{aligned}$$

Combining these observations yields

$$\sqrt{\mathrm{Tr}[\mathrm{Cov}(\mu)]} \le \sqrt{\mathrm{Tr}[\mathrm{Cov}(\widehat{\mu})]} + \big(\mathbb{E}\|X-Y\|_2^2\big)^{1/2}.$$

By definition,

$$W_2(\mu,\widehat{\mu}) = \inf_{\substack{X\sim\mu,\, Y\sim\widehat{\mu} \\ \text{over all couplings}}} \big(\mathbb{E}\|X-Y\|_2^2\big)^{1/2}.$$

Since the coupling was arbitrary, taking the infimum over couplings and using $W_2(\mu,\widehat{\mu}) \le \varepsilon$ gives

$$\sqrt{\mathrm{Tr}[\mathrm{Cov}(\mu)]} \le \sqrt{\mathrm{Tr}[\mathrm{Cov}(\widehat{\mu})]} + W_2(\mu,\widehat{\mu}) \le \sqrt{\mathrm{Tr}[\mathrm{Cov}(\widehat{\mu})]} + \varepsilon.$$

Rearranging and using the nonnegativity of $\sqrt{\mathrm{Tr}[\mathrm{Cov}(\widehat{\mu})]}$, we obtain

$$\sqrt{\mathrm{Tr}[\mathrm{Cov}(\widehat{\mu})]} \ge \big(\sqrt{\mathrm{Tr}[\mathrm{Cov}(\mu)]} - \varepsilon\big)_+.$$

Finally, since $\widehat{\mu} = T_{\#}\gamma$, the trace bound in Theorem 4.2 implies

$$\big(\mathbb{E}_\gamma\|J_T\|_F^2\big)^{1/2} \ge \sqrt{\mathrm{Tr}[\mathrm{Cov}(\widehat{\mu})]} \ge \big(\sqrt{\mathrm{Tr}[\mathrm{Cov}(\mu)]} - \varepsilon\big)_+,$$

which proves the claim. □

S3. Proofs for Section 5. For a nonempty closed set $A \subseteq \mathbb{R}^m$, let

$$\mathrm{dist}(y, A) = \inf_{a\in A} \|y-a\|_2$$

denote the Euclidean distance from y to A.

We use the Gaussian isoperimetric inequality [8] and the coarea formula [22, Section 3.4]. Together, these imply that every bounded locally Lipschitz function $g : \mathbb{R}^d \to \mathbb{R}$ satisfies

$$\int_{\mathbb{R}^d} \|\nabla g\|_2 \, d\gamma \ge \int_{\mathbb{R}} I\big(\gamma(\{g > t\})\big)\, dt, \tag{S3.1}$$

where the integral on the left may be infinite. An explicit formulation appears in Caselles, Miranda Jr., and Novaga [11, equation (11)].

PROOF OF THEOREM 5.1. Define

$$f(z) = \min\{\operatorname{dist}(T(z), A), \Delta\}.$$

The truncated distance is 1-Lipschitz, so f is locally Lipschitz and $\|\nabla f\|_2 \le \|J_T\|_{\mathrm{op}}$ almost everywhere. Since $0 \le f \le \Delta$, its gradient vanishes at differentiability points where $f = 0$ or $f = \Delta$. Moreover, $\{0 < f < \Delta\} = T^{-1}(U)$. Consequently,

$$\|\nabla f\|_2 \le \mathbf{1}_{T^{-1}(U)} \|J_T\|_{\mathrm{op}} \quad \text{almost everywhere.} \tag{S3.2}$$

For $0 < t < \Delta$, we have

$$\{f > t\} = T^{-1}((A^t)^c), \qquad \gamma(\{f > t\}) = 1 - P(A^t).$$

Applying (S3.1) and using the symmetry of I, we obtain

$$\begin{aligned}
\int_{T^{-1}(U)} \|J_T\|_{\mathrm{op}} \, d\gamma &\ge \int_{\mathbb{R}^d} \|\nabla f\|_2 \, d\gamma \\
&\ge \int_0^\Delta I\big(\gamma(\{f > t\})\big) \, dt \\
&= \int_0^\Delta I\big(P(A^t)\big) \, dt.
\end{aligned}$$

This proves the first inequality.

The sets A and $(A^\Delta)^c$ are disjoint, so $b \le 1/2$. The inclusions $A \subseteq A^t \subseteq A^\Delta$ give

$$b \le P(A^t) \le 1 - b.$$

By symmetry and monotonicity of I on $[0, 1/2]$, we therefore have $I(P(A^t)) \ge I(b)$. Integrating over $0 < t < \Delta$ proves the second inequality. □

PROOF OF COROLLARY 5.2. The definition of total variation and the assumed probability bounds for μ give

$$\begin{aligned}
P(A) &\ge \mu(A) - \varepsilon \ge w_1 - \varepsilon, \\
P((A^\Delta)^c) &\ge \mu((A^\Delta)^c) - \varepsilon \ge w_2 - \varepsilon, \\
P(U) &\le \mu(U) + \varepsilon \le \delta + \varepsilon.
\end{aligned}$$

Thus the quantity b in Theorem 5.1 satisfies $b \ge w - \varepsilon > 0$. Applying Theorem 5.1 and using monotonicity of I on $[0, 1/2]$, we obtain

$$\int_{T^{-1}(U)} \|J_T\|_{\mathrm{op}} \, d\gamma \ge \Delta I(w - \varepsilon) > 0.$$

Together with $P(U) \le \delta + \varepsilon$, this proves bound (5.4).

The integral is strictly positive, so $T^{-1}(U)$ has positive Gaussian measure. Therefore

$$P(U) = \gamma(T^{-1}(U)) > 0.$$

For $\delta + \varepsilon > 0$ and $q > 1$, Hölder's inequality gives

$$\begin{aligned}
\left(\int_{T^{-1}(U)} \|J_T\|_{\mathrm{op}}^q \, d\gamma \right)^{1/q} &\ge \frac{\displaystyle\int_{T^{-1}(U)} \|J_T\|_{\mathrm{op}} \, d\gamma}{P(U)^{1-1/q}} \\
&\ge \frac{\Delta I(w - \varepsilon)}{(\delta + \varepsilon)^{1-1/q}}.
\end{aligned}$$

The global moment dominates the localized moment by nonnegativity. These inequalities also hold when the moment is infinite. □

PROOF OF PROPOSITION 5.3. Define

$$f(z) = \min\{\operatorname{dist}(T(z), A), \Delta\}.$$

Since ν has a density, the gradient estimate (S3.2) established in the proof of Theorem 5.1 holds ν-almost everywhere. Consequently,

$$\int_{T^{-1}(U)} \|J_T\|_{\mathrm{op}}\, d\nu \geq \mathbb{E}_\nu \|\nabla f\|_2.$$

The function f equals 0 on $T^{-1}(A)$ and Δ on $T^{-1}((A^\Delta)^c)$. Each of these sets has probability at least b. Therefore, for every $c \in \mathbb{R}$,

$$\mathbb{E}_\nu |f - c| \geq b|c| + b|\Delta - c| \geq b\Delta.$$

Applying the Cheeger inequality in (5.6) to the bounded locally Lipschitz function f gives

$$\mathbb{E}_\nu \|\nabla f\|_2 \geq h_\nu \inf_{c\in\mathbb{R}} \mathbb{E}_\nu |f - c| \geq h_\nu \Delta b.$$

Combining the gradient estimate (S3.2) with the Cheeger inequality (5.6) and

$$\inf_{c\in\mathbb{R}} \mathbb{E}_\nu |f - c| \geq b\Delta$$

proves the localized bound (5.7).

To prove the claims concerning approximation of the target distribution μ, assume

$$\mathrm{TV}(P, \mu) \leq \varepsilon < w = \min\{w_1, w_2\}.$$

Applying the definition of total variation to the three sets A, $(A^\Delta)^c$, and U, and using the assumed probability bounds for μ, yields

$$P(A) \geq \mu(A) - \varepsilon \geq w_1 - \varepsilon,$$
$$P((A^\Delta)^c) \geq \mu((A^\Delta)^c) - \varepsilon \geq w_2 - \varepsilon,$$
$$P(U) \leq \mu(U) + \varepsilon \leq \delta + \varepsilon.$$

Hence $b \geq w - \varepsilon > 0$, and the localized bound (5.7) gives

$$\int_{T^{-1}(U)} \|J_T\|_{\mathrm{op}}\, d\nu \geq h_\nu \Delta (w - \varepsilon) > 0.$$

The integral is strictly positive, so $T^{-1}(U)$ has positive ν-measure. Therefore

$$P(U) = \nu(T^{-1}(U)) > 0.$$

For $\delta + \varepsilon > 0$ and $q > 1$, Hölder's inequality gives

$$\left(\int_{T^{-1}(U)} \|J_T\|_{\mathrm{op}}^q\, d\nu \right)^{1/q} \geq \frac{\displaystyle\int_{T^{-1}(U)} \|J_T\|_{\mathrm{op}}\, d\nu}{P(U)^{1-1/q}}$$
$$\geq \frac{h_\nu \Delta (w - \varepsilon)}{(\delta + \varepsilon)^{1-1/q}}.$$

Nonnegativity gives the global moment bound, completing the proof. □

S3.1. *Calculations for Example 5.4.* We verify the claims in Example 5.4. Throughout, $a > 0$ is fixed and $0 < \sigma \le 1$. We first show that the target moments remain bounded as $\sigma \downarrow 0$. We then show that the probability between the modes tends to zero and apply Corollary 5.2. Finally, we verify the derivative formula for the monotone transport.

To bound the target moments, let B take the values -1 and 1 with equal probabilities, independently of a standard Gaussian random variable G. Then

$$X_\sigma = aB + \sigma G$$

has distribution μ_σ. Since $|B| = 1$ and $\sigma \le 1$,

$$|X_\sigma| \le a + |G|.$$

Therefore, for every fixed $r > 0$,

$$\sup_{0<\sigma\le 1} \mathbb{E}|X_\sigma|^r \le \mathbb{E}(a + |G|)^r < \infty.$$

The last expectation is finite because Gaussian random variables have finite moments of every order. Thus every fixed absolute moment of the target remains bounded as $\sigma \downarrow 0$.

To compute the probability assigned to the separating region and apply Corollary 5.2, take

$$A = (-\infty, -a/2], \qquad \Delta = a, \qquad U = (-a/2, a/2).$$

For the Gaussian component with mean $-a$,

$$\mathbb{P}(-a + \sigma G \in U) = \mathbb{P}\left(\frac{a}{2\sigma} < G < \frac{3a}{2\sigma}\right).$$

The component with mean a gives the same probability by symmetry. Since the mixture weights sum to one, the probability $\delta_\sigma = \mu_\sigma(U)$ is given by

$$\begin{aligned}\delta_\sigma &= \Phi\left(\frac{3a}{2\sigma}\right) - \Phi\left(\frac{a}{2\sigma}\right) \\ &\le 1 - \Phi\left(\frac{a}{2\sigma}\right) \longrightarrow 0.\end{aligned}$$

The limit follows because $a/(2\sigma) \to \infty$.

By symmetry and continuity of the target distribution, each side of the valley has probability

$$w_\sigma = \frac{1 - \delta_\sigma}{2} \longrightarrow \frac{1}{2}.$$

For any locally Lipschitz exact transport T from a standard Gaussian source to μ_σ, Corollary 5.2 applies with $\varepsilon = 0$ and gives

$$\left(\mathbb{E}_\gamma \|J_T\|_{\mathrm{op}}^q\right)^{1/q} \ge \frac{aI(w_\sigma)}{\delta_\sigma^{1-1/q}}, \qquad q > 1.$$

For every fixed $q > 1$, the numerator tends to $aI(1/2) > 0$, while the denominator tends to zero. Hence the lower bound diverges, even though every fixed target moment remains bounded.

Finally, consider the univariate monotone transport

$$T_\sigma = F_\sigma^{-1} \circ \Phi,$$

where F_σ and f_σ are the cumulative distribution function and density of μ_σ. The density is smooth and strictly positive, so F_σ has a smooth inverse on $(0, 1)$. Consequently, T_σ is a locally Lipschitz exact transport.

Differentiating $F_\sigma(T_\sigma(z)) = \Phi(z)$ gives

$$f_\sigma(T_\sigma(z))T'_\sigma(z) = \varphi(z),$$

and therefore

$$T'_\sigma(z) = \frac{\varphi(z)}{f_\sigma(T_\sigma(z))}.$$

Symmetry gives $F_\sigma(0) = 1/2$, so $T_\sigma(0) = 0$. The target density at zero is

$$\begin{aligned} f_\sigma(0) &= \frac{1}{2\sigma}\varphi(a/\sigma) + \frac{1}{2\sigma}\varphi(-a/\sigma) \\ &= \frac{1}{\sigma}\varphi(a/\sigma). \end{aligned}$$

Substituting into the derivative formula yields

$$T'_\sigma(0) = \frac{\sigma\varphi(0)}{\varphi(a/\sigma)} = \sigma \exp\left(\frac{a^2}{2\sigma^2}\right) \longrightarrow \infty.$$

This verifies the derivative formula in Example 5.4.

S4. Proofs for Section 7.

S4.1. *Proofs for Bijective Reparametrizations.*

PROOF OF PROPOSITION 7.1. We first establish the exponential tail rate and then use it to prove the moment claims.

Since ψ_s is odd and strictly increasing, its inverse gives

$$\mathbb{P}(|Y| > y) = \mathbb{P}(|X| > s\sinh y) = (s\sinh y)^{-\alpha}L(s\sinh y)$$

for all sufficiently large y. Taking logarithms and dividing by y, we obtain

$$\frac{-\log \mathbb{P}(|Y| > y)}{y} = \alpha\frac{\log(s\sinh y)}{y} - \frac{\log L(s\sinh y)}{y}.$$

To evaluate the first term, use

$$s\sinh y = \frac{s}{2}(e^y - e^{-y}) = \frac{s}{2}e^y(1 - e^{-2y}), \qquad y > 0.$$

It follows that

$$\log(s\sinh y) = y + \log(s/2) + \log(1 - e^{-2y}).$$

We therefore obtain

$$\frac{\log(s\sinh y)}{y} \longrightarrow 1 \qquad \text{as } y \to \infty.$$

A positive slowly varying function satisfies

$$\lim_{x\to\infty} \frac{\log L(x)}{\log x} = 0$$

[5]. Since $s\sinh y \to \infty$ as $y \to \infty$, we consequently have

$$\frac{\log L(s\sinh y)}{y} = \frac{\log L(s\sinh y)}{\log(s\sinh y)}\frac{\log(s\sinh y)}{y} \longrightarrow 0 \qquad \text{as } y \to \infty.$$

Substituting these two limits proves the tail-rate formula (7.3).

We next show that $\mathbb{E}e^{t|Y|} < \infty$ for $0 < t < \alpha$. Fix such a t and choose b with $t < b < \alpha$. The tail-rate limit (7.3) implies that there is a finite $y_0 \geq 0$ such that

$$\mathbb{P}(|Y| > y) \leq e^{-by}, \qquad y \geq y_0.$$

Using the equality

$$e^{t|Y|} = 1 + t \int_0^{|Y|} e^{ty}\, dy$$

and applying Tonelli's theorem, we obtain

$$\begin{aligned}\mathbb{E}e^{t|Y|} &= 1 + t \int_0^{\infty} e^{ty} \mathbb{P}(|Y| > y)\, dy \\ &\leq 1 + t \int_0^{y_0} e^{ty}\, dy + t \int_{y_0}^{\infty} e^{-(b-t)y}\, dy.\end{aligned}$$

In the last expression, the first integral is finite because its integrand is continuous on the bounded interval $[0, y_0]$. The second integral is also finite, since

$$\int_{y_0}^{\infty} e^{-(b-t)y}\, dy = \frac{e^{-(b-t)y_0}}{b-t} < \infty.$$

Thus $\mathbb{E}e^{t|Y|} < \infty$.

Finally, to establish finiteness of every polynomial moment, fix $p > 0$ and choose any $t \in (0, \alpha)$. The function $u^p e^{-tu}$ is bounded on $[0, \infty)$, so there is a finite constant $M_{p,t}$ such that

$$u^p \leq M_{p,t} e^{tu}, \qquad u \geq 0.$$

It follows that

$$\mathbb{E}|Y|^p \leq M_{p,t} \mathbb{E}e^{t|Y|} < \infty.$$

This proves the polynomial-moment claim. □

S4.2. *Calculations for Example 7.2.* We verify the transformed density in Example 7.2 and prove the finite expected squared derivative statement (7.5). The calculation concerns a fixed $0 < \kappa \leq 2$.

Write p_κ for the Student-t density. The inverse transformation and its derivative are

$$x = \sqrt{\kappa} \sinh y, \qquad \frac{dx}{dy} = \sqrt{\kappa} \cosh y > 0.$$

Since this transformation is strictly increasing,

$$F_{\tau_\kappa}(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(X \leq \sqrt{\kappa} \sinh y).$$

Differentiating with respect to y and substituting the Student-t density gives

$$\begin{aligned}
f_\kappa(y) &= p_\kappa(\sqrt{\kappa}\sinh y)\sqrt{\kappa}\cosh y \\
&= \frac{\Gamma((\kappa+1)/2)}{\sqrt{\kappa\pi}\,\Gamma(\kappa/2)}\left(1+\frac{(\sqrt{\kappa}\sinh y)^2}{\kappa}\right)^{-(\kappa+1)/2}\sqrt{\kappa}\cosh y \\
&= \frac{\Gamma((\kappa+1)/2)}{\sqrt{\pi}\,\Gamma(\kappa/2)}(1+\sinh^2 y)^{-(\kappa+1)/2}\cosh y \\
&= \frac{\Gamma((\kappa+1)/2)}{\sqrt{\pi}\,\Gamma(\kappa/2)}(\cosh^2 y)^{-(\kappa+1)/2}\cosh y \\
&= \frac{\Gamma((\kappa+1)/2)}{\sqrt{\pi}\,\Gamma(\kappa/2)}(\cosh y)^{-(\kappa+1)}\cosh y \\
&= \frac{\Gamma((\kappa+1)/2)}{\sqrt{\pi}\,\Gamma(\kappa/2)}(\cosh y)^{-\kappa}.
\end{aligned}$$

Here the factors $\sqrt{\kappa}$ cancel, and we use $1+\sinh^2 y = \cosh^2 y$. Since $\cosh y > 0$, the remaining powers combine as

$$(\cosh^2 y)^{-(\kappa+1)/2}\cosh y = (\cosh y)^{-(\kappa+1)}\cosh y = (\cosh y)^{-\kappa}.$$

This proves (7.4).

To bound the derivative of the monotone transport map, we first compare the upper-tail probability of τ_κ with its density. Write

$$f_\kappa(y) = c_\kappa(\cosh y)^{-\kappa}, \qquad Q_\kappa(y) = 1 - F_{\tau_\kappa}(y) = \int_y^\infty f_\kappa(v)\,dv,$$

where $c_\kappa > 0$ is the normalizing constant in (7.4). For $y \geq 0$, the inequalities

$$\frac{e^y}{2} \leq \cosh y \leq e^y$$

give, upon taking reciprocals,

$$e^{-y} \leq \frac{1}{\cosh y} \leq 2e^{-y}.$$

Raising each term to the positive power κ and multiplying by c_κ therefore yields

$$c_\kappa e^{-\kappa y} \leq f_\kappa(y) \leq 2^\kappa c_\kappa e^{-\kappa y}, \qquad y \geq 0.$$

Integrating the upper bound over $[y,\infty)$ gives

$$Q_\kappa(y) \leq \frac{2^\kappa c_\kappa}{\kappa}e^{-\kappa y}.$$

Dividing by the lower bound on $f_\kappa(y)$, we obtain

$$\frac{Q_\kappa(y)}{f_\kappa(y)} \leq \frac{2^\kappa}{\kappa}, \qquad y \geq 0.$$

We now apply this bound to $S_\kappa = F_{\tau_\kappa}^{-1} \circ \Phi$. Let φ denote the standard Gaussian density. Since f_κ is smooth and strictly positive, S_κ is smooth, and differentiating

$$F_{\tau_\kappa}(S_\kappa(z)) = \Phi(z)$$

gives

$$S'_\kappa(z) = \frac{\varphi(z)}{f_\kappa(S_\kappa(z))}.$$

Both φ and f_κ are even. We first express this property in terms of their CDFs. For the transformed target, substituting $u = -v$ gives

$$\begin{aligned} F_{\tau_\kappa}(-y) &= \int_{-\infty}^{-y} f_\kappa(v)\, dv \\ &= \int_{y}^{\infty} f_\kappa(-u)\, du \\ &= \int_{y}^{\infty} f_\kappa(u)\, du \\ &= 1 - F_{\tau_\kappa}(y). \end{aligned}$$

The same argument for the even Gaussian density gives

$$\Phi(-z) = 1 - \Phi(z).$$

Using $F_{\tau_\kappa}(S_\kappa(z)) = \Phi(z)$, we therefore obtain

$$\begin{aligned} F_{\tau_\kappa}(S_\kappa(-z)) &= \Phi(-z) \\ &= 1 - \Phi(z) \\ &= 1 - F_{\tau_\kappa}(S_\kappa(z)) \\ &= F_{\tau_\kappa}(-S_\kappa(z)). \end{aligned}$$

Since $f_\kappa > 0$, the CDF F_{τ_κ} is strictly increasing. Equality of these CDF values therefore implies

$$S_\kappa(-z) = -S_\kappa(z),$$

so the monotone transport map S_κ is odd and its derivative S'_κ is even.

Oddness also gives $S_\kappa(0) = 0$. Moreover, $S_\kappa = F^{-1}_{\tau_\kappa} \circ \Phi$ is strictly increasing because both $F^{-1}_{\tau_\kappa}$ and Φ are strictly increasing. Consequently,

$$S_\kappa(z) > S_\kappa(0) = 0, \qquad z > 0.$$

In particular, $S_\kappa(z) \geq 0$ for $z \geq 0$, so we can apply the bound on $Q_\kappa(y)/f_\kappa(y)$ at $y = S_\kappa(z)$. To do so, we also use

$$Q_\kappa(S_\kappa(z)) = 1 - F_{\tau_\kappa}(S_\kappa(z)) = 1 - \Phi(z).$$

Consequently, for $z \geq 0$,

$$\begin{aligned} S'_\kappa(z) &= \frac{\varphi(z)}{1 - \Phi(z)} \frac{Q_\kappa(S_\kappa(z))}{f_\kappa(S_\kappa(z))} \\ &\leq \frac{2^\kappa}{\kappa} \frac{\varphi(z)}{1 - \Phi(z)}. \end{aligned}$$

A standard lower bound for the Gaussian Mills ratio is

$$\frac{1 - \Phi(z)}{\varphi(z)} \geq \frac{z}{1 + z^2}, \qquad z > 0.$$

Taking reciprocals, for $z \geq 1$ we obtain

$$\frac{\varphi(z)}{1-\Phi(z)} \leq z + \frac{1}{z} \leq 2z.$$

For $0 \leq z \leq 1$, we instead use $\varphi(z) \leq \varphi(0)$ and $1 - \Phi(z) \geq 1 - \Phi(1) > 0$, which give

$$\frac{\varphi(z)}{1-\Phi(z)} \leq \frac{\varphi(0)}{1-\Phi(1)} < \infty.$$

Combining the bounds for these two ranges, and recalling that κ is fixed, we obtain a finite constant C_κ such that

$$S'_\kappa(z) \leq C_\kappa(1+z), \qquad z \geq 0.$$

Since S'_κ is even, this bound extends to

$$S'_\kappa(z) \leq C_\kappa(1+|z|), \qquad z \in \mathbb{R}.$$

For $Z \sim N(0,1)$, squaring and taking expectation gives

$$\mathbb{E}[(S'_\kappa(Z))^2] \leq C_\kappa^2 \mathbb{E}[(1+|Z|)^2] < \infty.$$

This proves (7.5).

S4.3. *Reparametrizations for Multimodal Target Densities.* We provide the details for the reparametrization in Section 7.2. We first verify that Ψ is an invertible change of coordinates. We then prove the derivative comparisons in Propositions 7.3 and 7.4.

PROPOSITION S4.1 (Invertibility). *The map Ψ in (7.6) is a bijection from $\mathbb{R}$ onto $\mathbb{R}$ such that both Ψ and Ψ^{-1} are C^1, and satisfies $0 < \Psi' \leq 1$.*

PROOF. The distance function

$$d(x) = \min_i |x - c_i|$$

is continuous. The formula for a on $[L_\Psi, U_\Psi]$ in (7.7) is therefore continuous. At the endpoints $L_\Psi = c_1 - r$ and $U_\Psi = c_K + r$, the value of d is r, so this formula gives $a = 1$. Since $a(x) = 1$ for $x < L_\Psi$ and $x > U_\Psi$, the function a is continuous at both endpoints and hence on $\mathbb{R}$. Its definition also gives

$$0 < a(x) \leq 1, \qquad x \in \mathbb{R}.$$

By the fundamental theorem of calculus,

$$\Psi(x) = \int_0^x a(t)\,dt$$

is C^1 with $\Psi' = a > 0$. Thus Ψ is strictly increasing. Outside $[L_\Psi, U_\Psi]$, its derivative equals one, so $\Psi(x)$ equals x plus a constant on each tail. Consequently,

$$\lim_{x\to-\infty} \Psi(x) = -\infty, \qquad \lim_{x\to\infty} \Psi(x) = \infty.$$

Together with continuity and strict monotonicity, these limits show that Ψ is a bijection onto $\mathbb{R}$.

The inverse function theorem gives

$$(\Psi^{-1})'(y) = \frac{1}{a(\Psi^{-1}(y))}.$$

The right-hand side is continuous because a is continuous and strictly positive. Hence Ψ^{-1} is also C^1. □

Having established invertibility, we now compare the monotone transport maps from the standard Gaussian to the original and transformed targets. Proposition 7.3 gives a derivative bound for the map to the transformed target that is independent of the component separation D. Proposition 7.4 shows that the expected squared derivative of the map to the original target grows rapidly with D.

Let μ_D have density

$$p_D(x) = \tfrac{1}{2}\varphi(x + D/2) + \tfrac{1}{2}\varphi(x - D/2), \qquad D > 0, \tag{S4.1}$$

where φ and Φ are the standard Gaussian density and CDF. Write

$$m_1 = -D/2, \qquad m_2 = D/2$$

for the component means, and let F_D be the CDF of μ_D.

For the bound in Proposition 7.3, fix $r > 0$, $0 < h < 1$, and $0 \le \eta \le r$. Suppose the reference component locations satisfy

$$c_1 < c_2, \qquad |c_i - m_i| \le \eta, \quad i = 1, 2.$$

Construct Ψ using (7.6) and (7.7), and define

$$T_D(z) = F_D^{-1}(\Phi(z)), \qquad S_D(z) = \Psi(T_D(z)). \tag{S4.2}$$

For $Z \sim N(0,1)$, the variable $T_D(Z)$ has distribution μ_D, while $S_D(Z)$ has distribution $\Psi_{\#}\mu_D$. The CDF F_D specifies the exact transports used in this comparison. Constructing Ψ requires only the reference locations and the parameters r, h.

To obtain formulas for the transport derivatives, differentiate

$$F_D(T_D(z)) = \Phi(z).$$

Since p_D is smooth and strictly positive, this gives

$$p_D(T_D(z))T_D'(z) = \varphi(z).$$

Let

$$I(u) = \varphi(\Phi^{-1}(u)), \qquad 0 < u < 1.$$

For $x = T_D(z)$, we have $z = \Phi^{-1}(F_D(x))$. Applying the chain rule to $S_D = \Psi \circ T_D$ therefore yields

$$T_D'(z) = \frac{I(F_D(x))}{p_D(x)}, \qquad S_D'(z) = a(x)\frac{I(F_D(x))}{p_D(x)}. \tag{S4.3}$$

These formulas identify the factor $a(x)$ through which the change of coordinates modifies the transport derivative.

We will use the elementary properties

$$I(u) \le \frac{1}{\sqrt{2\pi}}, \qquad I(1-u) = I(u), \qquad I'(u) = -\Phi^{-1}(u).$$

In particular, I is increasing on $(0, 1/2]$ and decreasing on $[1/2, 1)$.

PROOF OF PROPOSITION 7.3. Fix $z \in \mathbb{R}$ and write $x = T_D(z)$. We bound $S_D'(z)$ separately inside and outside $[L_\Psi, U_\Psi]$. Inside this interval, the factor $a(x)$ offsets the small mixture density in the denominator of (S4.3). Outside the interval, $a(x) = 1$, and we bound the original transport derivative directly.

First suppose $x \in [L_\Psi, U_\Psi]$. Choose a nearest reference location c_j, so that $|x - c_j| = d(x)$. The reference-location error bound gives

$$|x - m_j| \le |x - c_j| + |c_j - m_j| \le d(x) + \eta.$$

The corresponding Gaussian component therefore gives

$$p_D(x) \ge \tfrac{1}{2}\varphi(x - m_j) \ge \frac{1}{2\sqrt{2\pi}} \exp\left(-\frac{(d(x)+\eta)^2}{2}\right).$$

Combining this bound with $I \le 1/\sqrt{2\pi}$, (S4.3), and the formula for a in (7.7), we obtain

$$S_D'(z) \le 2 \exp\left(\frac{(d(x)+\eta)^2}{2} - \frac{(d(x)-r)_+^2}{2h^2}\right).$$

We next bound the exponent independently of $d(x)$. If $d(x) \le r$, then

$$\frac{(d(x)+\eta)^2}{2} - \frac{(d(x)-r)_+^2}{2h^2} \le \frac{(r+\eta)^2}{2} \le \frac{(r+\eta)^2}{2(1-h^2)}.$$

If $d(x) > r$, write $d(x) = r + u$ with $u > 0$. Completing the square gives

$$\begin{aligned}\frac{(u+r+\eta)^2}{2} - \frac{u^2}{2h^2} &= \frac{(r+\eta)^2}{2(1-h^2)} \\ &\quad - \frac{1-h^2}{2h^2}\left(u - \frac{h^2(r+\eta)}{1-h^2}\right)^2 \\ &\le \frac{(r+\eta)^2}{2(1-h^2)},\end{aligned}$$

where the inequality uses $0 < h < 1$. Both cases therefore give

$$S_D'(z) \le 2 \exp\left(\frac{(r+\eta)^2}{2(1-h^2)}\right) =: C$$

for $x \in [L_\Psi, U_\Psi]$. This constant depends only on r, h, η and is independent of D.

Now suppose $x < L_\Psi = c_1 - r$. Since $c_1 - m_1 \le \eta \le r$, we have $L_\Psi \le m_1 < m_2$. Thus x lies to the left of both component means, and

$$F_D(x) = \tfrac{1}{2}\Phi(x - m_1) + \tfrac{1}{2}\Phi(x - m_2) \le \Phi(x - m_1) \le \tfrac{1}{2}.$$

Because I is increasing on $(0, 1/2]$,

$$I(F_D(x)) \le I(\Phi(x - m_1)) = \varphi(x - m_1).$$

Together with $p_D(x) \ge \varphi(x - m_1)/2$ and $a(x) = 1$, the formula for S_D' in (S4.3) gives

$$S_D'(z) \le 2 \le C.$$

Similarly, suppose $x > U_\Psi = c_2 + r$. Since $m_2 - c_2 \le \eta \le r$, we have $m_1 < m_2 \le U_\Psi < x$. Consequently,

$$F_D(x) \ge \Phi(x - m_2) \ge \tfrac{1}{2}.$$

Because I is decreasing on $[1/2, 1)$,

$$I(F_D(x)) \le I(\Phi(x - m_2)) = \varphi(x - m_2).$$

Using $p_D(x) \ge \varphi(x - m_2)/2$ and $a(x) = 1$ in (S4.3) again yields

$$S_D'(z) \le 2 \le C.$$

All factors in the expression for S'_D in (S4.3) are strictly positive. Combining the three regions proves

$$0 < S'_D(z) \le C, \qquad z \in \mathbb{R}.$$

Squaring and taking expectation gives

$$\mathbb{E}[S'_D(Z)^2] \le C^2, \qquad Z \sim N(0,1),$$

as required. □

We next prove the growth bound for the original monotone transport map T_D. Near the midpoint between the component means, the density $p_D(x)$ becomes exponentially small as D increases, while $I(F_D(x))$ remains bounded below. We use an interval around this midpoint to obtain a lower bound on the expected squared derivative.

PROOF OF PROPOSITION 7.4. Since $T_D(Z)$ has density p_D, the formula for T'_D in (S4.3) gives

$$\begin{aligned}\mathbb{E}[T'_D(Z)^2] &= \int_{\mathbb{R}} \left(\frac{I(F_D(x))}{p_D(x)}\right)^2 p_D(x)\,dx \\ &= \int_{\mathbb{R}} \frac{I(F_D(x))^2}{p_D(x)}\,dx,\end{aligned}$$

allowing either side to be infinite. We obtain a lower bound by restricting the last integral to $[-1/D, 1/D]$.

For $D \ge 2$, this interval lies between the component means. Thus, for every x in the interval,

$$\frac{1}{2} \le \Phi(x - m_1) \le 1, \qquad 0 \le \Phi(x - m_2) \le \frac{1}{2}.$$

Averaging the two component CDFs gives

$$\frac{1}{4} \le F_D(x) \le \frac{3}{4}.$$

The symmetry and monotonicity of I therefore imply

$$I(F_D(x)) \ge I(1/4) > 0.$$

On the same interval, the distance from x to either component mean is at least $D/2 - 1/D$. Since the standard Gaussian density decreases with distance from zero,

$$\begin{aligned}p_D(x) &\le \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{D}{2} - \frac{1}{D}\right)^2\right] \\ &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{D^2}{8} + \frac{1}{2} - \frac{1}{2D^2}\right) \\ &\le \frac{e^{1/2}}{\sqrt{2\pi}} e^{-D^2/8}.\end{aligned}$$

Combining the lower bound on $I(F_D(x))$ with the upper bound on $p_D(x)$, and using the interval length $2/D$, we obtain

$$\begin{aligned}\mathbb{E}[T'_D(Z)^2] &\ge \int_{-1/D}^{1/D} \frac{I(F_D(x))^2}{p_D(x)}\,dx \\ &\ge \frac{2}{D}\, I(1/4)^2 \sqrt{2\pi}\, e^{-1/2} e^{D^2/8} \\ &= \frac{c}{D} e^{D^2/8},\end{aligned}$$

where $c = 2I(1/4)^2\sqrt{2\pi}\, e^{-1/2} > 0$ is independent of D. This proves the lower bound (7.9). □

S5. Experimental Details and Additional Experiments.

S5.1. *Details of the Heavy-Tail Illustrations.* These distribution families are widely used in many fields such as climatology, hydrology, and finance [9, 16, 35], making them natural and representative test cases. They are also particularly well-suited for illustrating our theory for several reasons. First, a transport map from a Gaussian source can be constructed explicitly for these targets, enabling direct inspection of their gradients and Jacobians. This allows us to numerically assess the tightness and practical relevance of the theoretical lower bounds derived in Section 4. Second, the tail behavior of these families is governed by interpretable parameters (i.e., the degrees of freedom for the Student-t family and the shape parameters for the Pareto and Fréchet families), providing explicit control over moment growth and allowing us to systematically investigate varying degrees of tail heaviness.

In the multivariate case, when pushing forward a standard Gaussian in $\mathbb{R}^d$ to a product distribution also in $\mathbb{R}^d$, the pushforward map can be applied coordinate-wise. If $d = 2$, this is given by

$$T(z_1, z_2) = \big(F_\mu^{-1}(F_\nu(z_1)), F_\mu^{-1}(F_\nu(z_2))\big).$$

The Jacobian of T is thus also diagonal. We adopt this construction for the explicit heavy-tail simulations below.

As target distributions, we consider three canonical families of heavy-tailed laws: (a) Student-t distributions, (b) Pareto distributions, and (c) Fréchet distributions. Here, all target measures are product measures with independent coordinates. For ease of visualization, we present results in the two-dimensional setting. Since all target distributions are product measures, this entails no loss of generality, and the behavior observed in two dimensions is representative of higher-dimensional cases.

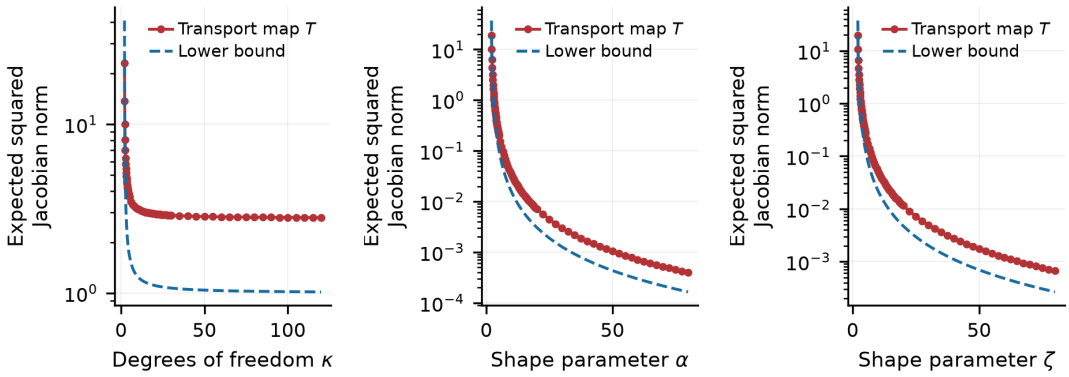
For each parameter setting, we plot both sides of the trace bound Poincaré inequality given in part (iii) of Theorem 4.2. Specifically, the lower bound (left-hand side of the inequality) corresponds to the trace of the covariance matrix of the target distribution, which admits a closed-form expression for the distributions considered. The right-hand side of the inequality is given by the sum of the expected squared norm of the Jacobian diagonals of the transport map under the Gaussian source measure. We compute this expectation numerically using Gauss–Hermite quadrature. Plotting these quantities together allows us to directly assess the tightness and practical relevance of our theoretical bounds.

S5.2. *Empirical Illustrations with a Log-Concave Source Distribution.* The simulations in Section 8 take the source distribution to be a standard Gaussian, since this is the most common choice in pushforward generative modeling. Our theory in Section 4, however, applies to any isotropic log-concave source distribution. Here we replicate the experiments of Section 8 with the Gaussian source replaced by a Laplace distribution, which is log-concave but not strongly log-concave. More precisely, the Laplace distribution under consideration is the product of two independent Laplace distributions with location parameter 0 and scale parameter $1/\sqrt{2}$ (since we want its variance to be 1 to satisfy the isotropic condition). The target families and overall procedure are otherwise identical to those of Section 8: we consider the Student-t, Pareto, and Fréchet families with parameters controlling tail heaviness and construct the transport map coordinate-wise via the probability integral transform $T(z) = F_\mu^{-1}(F_\nu(z))$ as in Section 8.1.

We report (a) comparisons of the numerically evaluated expected squared Frobenius norm of the Jacobian against the lower bound predicted by the log-concave Poincaré inequality

(Theorem 4.5, part (iii) of Theorem 4.2), and (b) heatmaps of the log absolute Jacobian determinant across the input space. For the Gaussian case of the Poincaré inequality, the leading constant is known. For log-concave distributions, the constant in the Poincaré inequality is in general not known. Thus, we use a working constant of 2 for our experiments here, which allows us to inspect how closely the bound tracks the Jacobian size as tail behavior changes over these families.

The purpose of these supplementary experiments is to verify that the qualitative behavior identified in the Gaussian case is not an artifact of the Gaussian source: heavy-tailed targets continue to force large Jacobian norms when transporting from a non-Gaussian, log-concave source, in agreement with the moment-to-Jacobian inequalities for general isotropic log-concave measures.

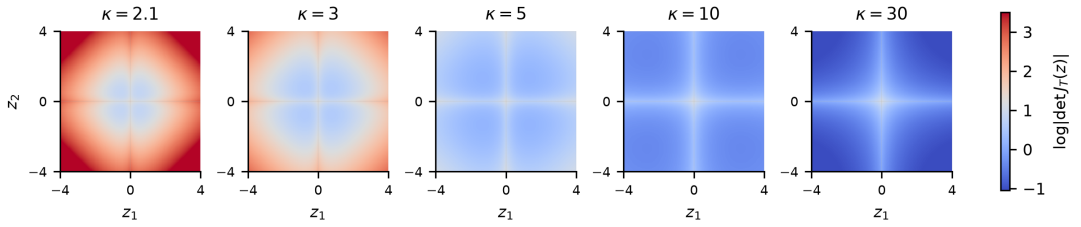


(a) Student-t target (b) Pareto target (c) Fréchet target

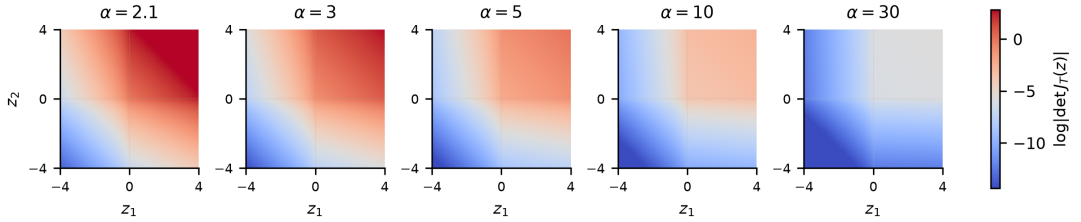
Fig S1: Comparison of Poincaré lower bounds with the expected squared Frobenius norm of the transport Jacobian for a product Laplace source and heavy-tailed target distributions. The red curves with circular markers show the numerically evaluated quantities. The blue dashed curves show the lower bounds, using $C_{\mathrm{PI}}(\nu) = 2$.

As shown in Figure S1, the numerical estimates of the expected squared Frobenius norm continue to track the log-concave Poincaré lower bounds across all three target families and across the full range of tail-heaviness parameters considered. The heatmaps in Figure S2 also display the same qualitative pattern as in the Gaussian-source case (Figure 3): rapid Jacobian growth in regions corresponding to the heavy-tailed behavior of the target. Together, these results confirm that the moment-to-Jacobian phenomenon identified in the main text is robust to the choice of (light-tailed) log-concave source and is not specific to Gaussian sources.

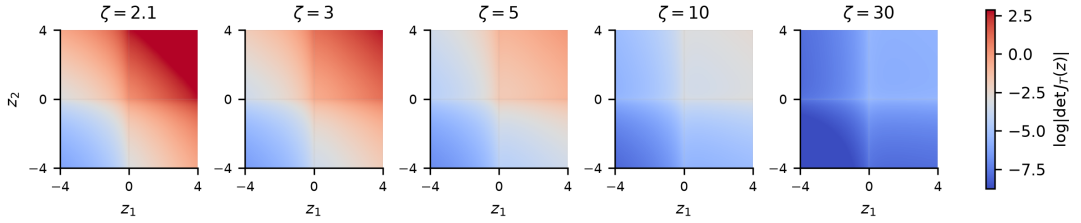
S5.3. *Empirical Illustrations with Correlated Heavy-Tailed Targets.* The simulations in Section 8 take the target distribution to be a product measure with independent coordinates, since the resulting transport map is then available coordinate-wise in closed form via the probability integral transform. Our theoretical results in Section 4, however, are stated in full generality and do not require the target to be a product measure. In this section, we replicate the experiments of Section 8 with the independent product target replaced by a correlated two-dimensional heavy-tailed targets. We replicate the experiments with correlated heavy-tailed targets obtained as linear transformations of independent heavy-tailed coordinates. Specifically, we first construct an independent heavy-tailed vector with Student-t, Pareto, or



(a) Student-t target



(b) Pareto target



(c) Fréchet target

Fig S2: Log absolute Jacobian determinant heatmaps for log-concave-to-heavy-tailed transport maps. As in the Gaussian-source case (Figure 3), the magnitude of the Jacobian grows rapidly away from the center as the target becomes increasingly heavy-tailed. Note that the Pareto and Fréchet distributions are right-skewed and supported on the positive orthant.

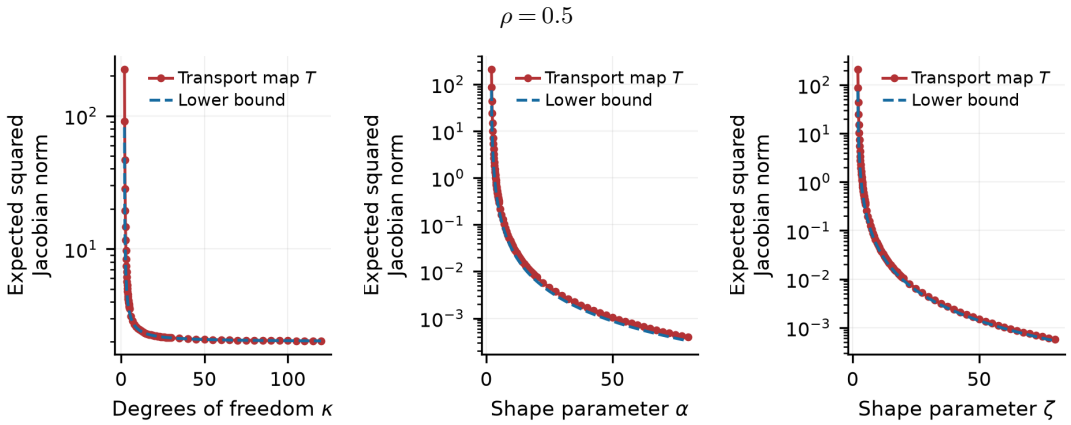
Fréchet coordinates as before, and then we apply a linear map L, chosen as the Cholesky factor of a prescribed correlation matrix.

The procedure is otherwise identical to that of Section 8: we transport a standard Gaussian source to the correlated heavy-tailed target, vary the parameters that control tail heaviness and the correlation parameter $\rho \in \{0.5, 0.9\}$, and report comparisons of the numerically evaluated expected squared Frobenius norm of the Jacobian against the lower bound predicted by the Gaussian Poincaré inequality (Theorem 4.2, part (iii)).

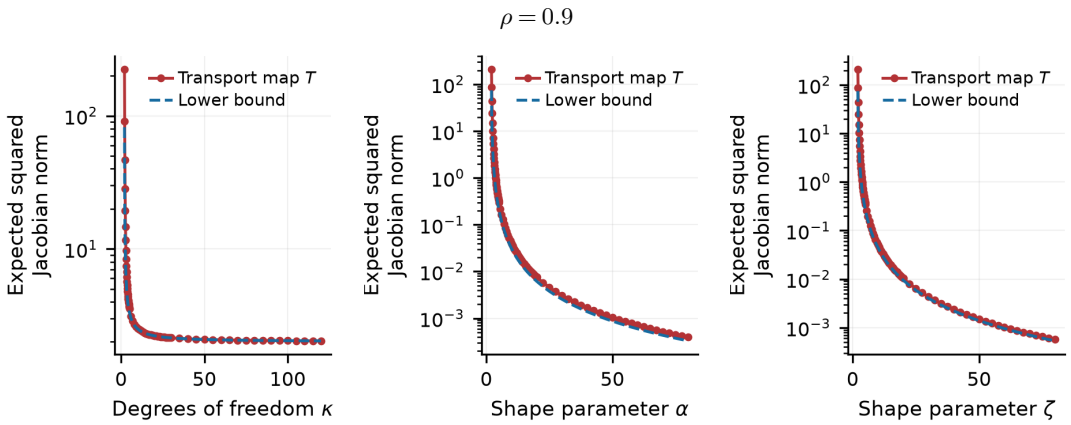
These supplementary experiments verify that the qualitative behavior identified in the product-target case extends to targets with nontrivial dependence structure. Heavy-tailed correlated targets continue to force large Jacobian norms, in agreement with our inequalities, which apply to general (not necessarily product) targets.

As shown in Figure S3, the numerical estimates of the expected squared Frobenius norm continue to closely track the Poincaré-based lower bounds across all three target families when the independent heavy-tailed coordinates are linearly transformed to introduce correlation.

S5.4. *Details of Numerical Experiments.* For the heavy-tail experiment in Figure 5, we compare the exact maps T and $S = \Psi \circ T$ using a standard Gaussian source and no additional standardization. For the two-dimensional product Student, Pareto, and Fréchet targets, we apply



(A) Student-t target (B) Pareto target (C) Fréchet target



(D) Student-t target (E) Pareto target (F) Fréchet target

Fig S3: Comparison of Poincaré lower bounds with the expected squared Frobenius norm of the transport Jacobian for correlated heavy-tailed targets under a standard Gaussian source. The top row, panels (A)–(C), uses $\rho = 0.5$, and the bottom row, panels (D)–(F), uses $\rho = 0.9$. Columns correspond to the Student-t, Pareto, and Fréchet target constructions. Each panel shows the numerically evaluated quantity in red and the lower bound computed from the trace of the target covariance matrix as a blue dashed curve. Compare with Figure 2, where the target has independent coordinates.

$\operatorname{arsinh}(x/s)$ coordinate-wise, with $s = \sqrt{\kappa}$ for Student-t and $s = 1$ otherwise. Expectations are evaluated using Gauss–Legendre quadrature.

For the multimodal experiment in Figure 6, we use five equally weighted Gaussian components with unit standard deviations and means

$$m_i = D(i-3), \quad i = 1, \ldots, 5, \qquad D \in \{3, 4, \ldots, 16\}.$$

The supplied reference component locations have the following errors relative to the component means:

$$(c_1 - m_1, c_2 - m_2, c_3 - m_3, c_4 - m_4, c_5 - m_5) = (0.4, -0.3, 0.5, -0.4, 0.2).$$

These differences are held fixed for every value of D. Throughout the study, $r = 1.5$ and $h = 0.4$ are fixed. No target CDF, weights, or individual component widths are supplied to the transformation. The reference component locations are supplied synthetically to isolate the reparametrization; this study does not evaluate a procedure for locating components.

For both $g = T$ and $g = S$, we report $\mathbb{E}[g'(Z)^2]$ for $Z \sim N(0,1)$. Writing p, F for the mixture density and CDF, and $I(u) = \varphi(\Phi^{-1}(u))$, we evaluate

$$\mathbb{E}[S'(Z)^2] = \int_{\mathbb{R}} \frac{a(x)^2 I(F(x))^2}{p(x)}\, dx. \tag{S5.1}$$

Setting $a = 1$ gives the original-map quantity.

S5.5. *Density-Based Fitting for Figure 1.* The heavy-tailed target has two independent standard Student-t coordinates with 2.1 degrees of freedom. The bimodal target is $\frac{1}{2}N(m_1, \sigma^2 I_2) + \frac{1}{2}N(m_2, \sigma^2 I_2)$, where $m_1 = (-10, 0)$, $m_2 = (10, 0)$, and $\sigma = 0.65$. Both experiments use a standard Gaussian source on $\mathbb{R}^2$.

For the Student-t target, apply (7.2) coordinate-wise with $s = \sqrt{2.1}$. For the Gaussian mixture, apply (7.6) separately to each coordinate, using the reference component locations $c_i = m_i + 0.25\sigma v_i$, with $v_1 = (0.8, -0.6)$ and $v_2 = (-0.7, 0.9)$. Set $r = 1.5\sigma = 0.975$ and $h = 0.9\sigma = 0.585$.

The normalizing flows are trained by minimizes reverse Kullback-Leibler divergence against the original target density p. Writing T_θ for the complete generator, q_θ for its density, and ϕ_2 for the standard Gaussian density, the objective is

$$D_{\mathrm{KL}}(q_\theta \| p) = \mathbb{E}_{Z \sim N(0, I_2)}[\log \phi_2(Z) - \log |\det J_{T_\theta}(Z)| - \log p(T_\theta(Z))]\,.$$

The learned architecture is a four-layer affine-coupling RealNVP flow [17]. The network has two 32-unit $\tanh$ hidden layers. Training consists of 800 Adam updates with batch size 512 and learning rate 10^{-3}. Heatmaps are drawn with 100,000 generated samples. Target contours are evaluated analytically.